Artificial Intelligence models to enhance cognitive intervention in older adults with Subjective Cognitive Decline: pilot study

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Artificial Intelligence models to enhance cognitive intervention in older adults with Subjective Cognitive Decline: pilot study

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The undersigned Musteata Stela, student number ID 009701, born in Pitusca (Moldova) on 8th of October 1988, author of this thesis entitled "Predicting cognitive intervention outcome in older adults with subjective cognitive decline: a pilot study".

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INTRODUCTION

Subjective cognitive decline

Cognitive decline is not an inevitable fate for older adults, and some seniors maintain excellent cognitive function up to 70 or 80 years of age[1]. However, for others, cognitive decline can begin in their 50's or 60s. Generally, it is difficult to separate normal cognitive changes due to aging from those associated with pathological cognitive decline. For many people cognitive decline, a continuum of cognitive changes (e.g. processing speed, attention, working memory, cognitive flexibility, and episodic memory), is associated with relatively minor and sporadic difficulties that are considered normal within the spectrum of typical cognitive aging (e.g. Subjective Cognitive Decline: SCD)[2,3]. Sometimes these cognitive changes are serious enough to be noticed by other people, but the changes still do not interfere with daily life or independent function and thus is referred to as **M**ild **C**ognitive **I**mpairment (MCI). For others, cognitive decline is associated with severe cognitive deficits[4] that impede the ability to live independently (i.e., dementia).

Subjective **C**ognitive **D**ecline (SCD) refers to everyday concerns of seniors that their cognition is failing despite performing within normal limits on standardized neuropsychological tests and retaining normal adaptive functioning[5,6]. SCD was introduced in the early 1980s by Reisberg and colleagues (1982) and was considered a stage that preceded MCI, the symptomatic predementia phase of Alzheimer Disease (AD) or of other forms of dementia. Over the years, various terms were used with the same meaning as SCD: subjective memory complaints, subjective cognitive complaints, subjective memory concerns, subjective memory impairment, subjective cognitive impairment, etc. Besides all these terminologies, Jessen and colleagues (2014) defined specific inclusion and exclusion criteria for assessing SCD[7] (Figure 1).

Figure 1.Research criteria for SCD

Research criteria for SCD (Jessen et al., 2014)

Inclusion criteria (both points must be present):

1. self-reported persistent decline in cognitive functions compared with a previously normal status and unrelated to an acute event; and
2. the performance on standardized cognitive tests needs to be adjusted by age, gender, and education that are used to classify MCI or prodromal AD.

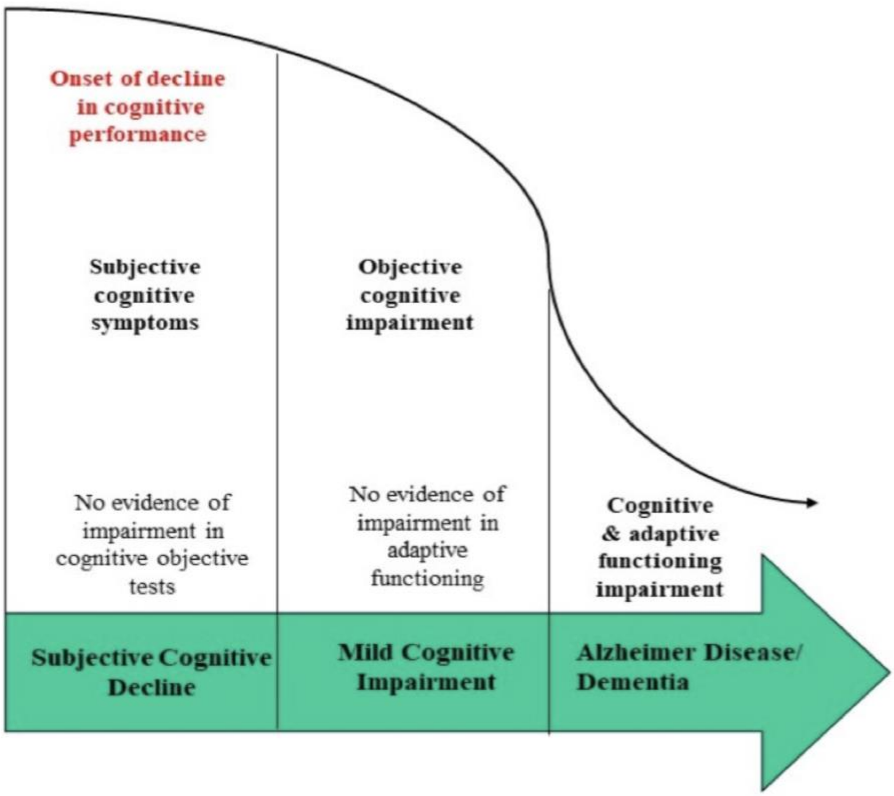
Exclusion criteria:

- Mild cognitive impairment, prodromal AD, or dementia
- Condition can be explained by a psychiatric disorder or neurological disease (apart from AD), medical disorder, medication, or substance use.

The majority of individuals with SCD never progress to significant cognitive decline, but some of them have a higher risk of progression to objective cognitive impairment than persons with no cognitive concerns. This kind of individuals constitute a heterogeneous group[8]. The great majority of older adults are ‘‘worried well’’ and don’t deteriorate more rapidly than usual (i.e. common aging decline or pseudodementia). In contrast, some participants with SCD have evidence of preclinical Alzheimer disease (AD) and, as such, a higher risk of progression to dementia.

The Figure 2 describes clearly the transition from SCD to MCI and from MCI to dementia or AD [7]. SCD is considered as a pre-MCI condition in which the cognitive performance on objective tests is within normal limits. Once subject's cognitive performance falls below the normal limits (scores adjusted by age, gender and education) retaining a normal adaptive functioning we can diagnose MCI. Lately, the cognitive decline progresses steadily toward dementia.

Figure 2. Progression of cognitive decline and clinical stages



SCD and biomarkers of preclinical AD

Numerous longitudinal studies[5,9–11] have suggested that SCD in older adults may be considered as an early indicator for a higher risk of developing Alzheimer or other forms of dementia.

Individuals with SCD show a pattern of brain atrophy in medial temporal and frontotemporal regions compared to clinically normal older adults[1213]. Several biomarker studies investigated the relationship between SCD and the medial temporal lobe. Specifically, reductions in bilateral entorhinal cortex volume and left hippocampal volume were observed in individuals with SCD compared to individuals without memory complaints[14–16]. Imaging studies have revealed that SCD is associated with decreased hippocampal volume (specifically in the CA1 and subiculum subfields)[17,18] which is in line with AD-pattern of gray matter atrophy and cortical thinning in the medial temporal lobe[19,20].

An imaging technique using Positron Emission Tomography (PET) with Pittsburgh Compound B (PiB) was adopted for the quantification of amyloid burden in vivo[21]. Using amyloid-PET was discovered that greater amyloid is associated with a greater number of memory complaints on questionnaires[5,22,23]. These findings are in contrast to other studies that have not found associations between amyloid and SCD[24–26]. Other studies used FDG Positron Emission Tomography (FDG-PET) to identify brain regions that show abnormal glucose metabolism in SCD. A decline of FDG uptake in posterior cingulate, temporoparietal, and prefrontal association cortex in subjects with MCI was found by several studies of Perani’s research group[13,27,28]. Other interesting findings showed reduced parietotemporal and parahippocampal glucose metabolism and increased metabolism in the right medial temporal lobe[20,29]. That being said, parietotemporal regions seems to be vulnerable in early AD, and hyperactivity may reflect compensation that precedes eventual atrophy.

SCD and objective cognition

Objective cognitive testing is considered the gold standard for assessment of cognitive efficiency or decline. Nevertheless, it is complicated to assess the cognitive decline in individuals with SCD when by definition their cognitive performance is within normal limits. Therefore, it is important to consider that neuropsychological tests assess performance at a single point in time, whereas subjective decline captures longitudinal change[7]. Furthermore, for some older adults with SCD, average scores at neuropsychological tests will reflect stable functioning (no decline), whereas for others average scores will reflect a subtle decline from a higher baseline level[5].

Minor cognitive difficulties are more difficult to differentiate from normal ability due to low test sensitivity and the adequacy of normative datasets for older adults[5,7]. Therefore, self-reported questionnaires are used in clinical and research field[30–32]. For example, the most used is Memory Complaint Questionnaire (MAC-Q): a self-report tool, developed with the intent of screening for Age-Associated Memory Impairment (AAMI) and for research that requires an index of memory complaint. The original validation of the MAC-Q was performed with participants who was reporting memory complaint with no evidence of cognitive decline or depression, and a score of 25 and above on the MAC-Q was considered indicative of memory decline[33,34]. Thus, it is sometimes difficult to determine if changes, at least in early stages, are within normal limits or pathological (e.g. normal aging is characterized by a decline in abilities such as episodic memory and executive functioning). In addition, age-related changes in thinking and remembering and dementia pathology can often occur simultaneously and can be very subtle[35]. A longitudinal study showed that seniors with SCD, particularly those with concerns about their memory, demonstrated steeper episodic memory decline longitudinally (i.e. at each 6 months follow-up), but not declines in verbal fluency or working memory[36]. In contrast, another study noticed that lower objective scores in executive function domains were associated with subsequent increases in memory complaints[5]. Another interesting finding for SCD subjects was the association between lower memory complaints and stronger memory performances (consistent with practice effects) or lower memory performance and decline in memory complaints (consistent with anosognosia). That being said, let's find out what kind of cognitive complaints may have older adults with SCD.

Cognitive complaints in older adults with SCD

Due to irregular changes in brain structure, cognitive changes in aging are not uniform across all cognitive domains or across all individuals. Below are described the cognitive areas that are most susceptible to aging decline.

Attention

Attention is a complex cognitive domain that has multiple subcomponents specialized in processing of various types of cognitive information. There are four important types of attention: selective (i.e. focus on one thing at a time), sustained (i.e. focus for a long period of time), divided (i.e. focus on two events at once), and alternating attention (i.e. switch the focus back and forth between tasks). Some of these types of attention are involved in all cognitive domains. Therefore, age-related changes in attention can have important effects on individual's function in daily life activities[37].

Older adults show significant impairments on tasks that require flexible control such as shifting between multiple tasks[38,39]. Furthermore, they are slower than young adults on tasks that require selective attention in an environment full of distractors[40]. Many seniors report impairments in tasks that require flexible control of attention (i.e. divided attention and alternating attention). In contrast, older adults are able to sustain attention for a long period of time. There are many evidences that divided and alternating attention appear to respond well to cognitive training[41–43].

Working memory

Working memory (WM) is a cognitive system for temporary storage and manage of the information required in many complex cognitive tasks such as learning, planning, problem-solving, decision-making and comprehension[44]. These complex cognitive tasks require continuous integration and reorganization of information from a variety of sources[45]. The brain region that is mostly active during working memory tasks is pre-frontal cortex (PFC) with left PFC involved in verbal tasks and right PFC in visuospatial tasks[46]. Several neuroimaging studies[47,48] have confirmed that PFC is the first area to decline with advancing age.

Prominent theories have been hypothesized WM being the central source of age-related cognitive deficits[38,44,49–51]. Salthouse has demonstrated in numerous aging studies that slowing of information processing is a primary cognitive deficit involved in many cognitive tasks, including working and long-term memory. A meta-analysis[52] on verbal memory span in aging has revealed that older adults show minimal or no deficits in short-term memory task. They can easily hold about 7 ± 2 digits in mind as long as the digits are being rehearsed. On working memory tasks, such as repeating the numbers backwards, older adults encounter more difficulties, because in these kinds of tasks the information has to be continuously manipulated and reorganized[52,53]. Hasher, Zacks, and May (1999), in their theory of cognitive aging, suggest that a lack of inhibitory control (i.e. the failure to suppress irrelevant information in a WM span task) contributes in many cognitive deficits.

Indeed, older adults exhibit significant deficits in higher-level cognitive tasks[40] that uses WM such as driving a car, financial management, decision-making, planning, etc.

Memory

Memory is a complex and multifaceted cognitive function that has probably received the most attention in aging[54]. How does this function work? The information is encoded in the present (i.e. short-term memory) or past (i.e. long-term memory) and is recalled through declarative and procedural processes. Declarative memory refers to the ability of consciously recall facts (i.e. semantic memory), events (i.e. episodic memory), or perceptual information (i.e. perceptual memory). Procedural memory requires the implicit recall of information and is usually divided into procedural, priming, or simple conditioning paradigms.

Like attention, some kinds of memory remain relatively intact with age while others show significant declines. Semantic memory is among the more stable memory systems across the adult life span whereas episodic memory declines considerably with age. For example, older adults with subjective memory complaints have more difficulty recalling what they had for breakfast (i.e. episodic memory) than recalling the names of common objects (i.e. semantic memory). More problematic for seniors is recalling context or source information: where or when something was heard, seen, or happened (i.e. reality monitoring). Encoding and retrieving all these context details may be particularly demanding of attentional resources not only of memory abilities. Problems with episodic memory may be also due to high levels of emotion or stress[55] and this is an important consideration that needs to be done.

Perception

The older adults may also complain having visual perception difficulties. It was observed that challenges with visual perception depend on the stimulus complexity (computational load or complexity of neural network). Any object in an image that is defined by luminance or colour is said to have a "first order" property. If an object is defined by depth, texture or motion (without any luminance or colour cues), it is said to have "second order" properties because more cortical processing levels are required to perceive the image[56]. The perceptual loss due to aging is more evident in the second-order motion and texture tasks as opposed to the first-order tasks[56].

Faubert J. (2002) showed that there was no effect of aging for colour-defined motion (a first-order attribute) when sensory input differences are compensated for. On the contrary, he noticed a dramatic effect of aging when size discrimination for two squares had to be done simultaneously or when spatial frequency information for two gratings had to be transferred from one attribute to another for comparison (e.g., colour to luminance). Both of these tasks are presumed to require more complex neural networks. Finally, visual working memory for low-level spatial information appears unaffected by aging when individual differences of sensory input are compensated for. Although the psychophysical response for a perceptual task with low-level stimuli may not be affected, it is not to say that there is no underlying change in the neural networks required to process such information in the older individuals. The functional imaging studies clearly demonstrate that changes have occurred in the neural circuitry involved but they may not have reached a point of saturation when simple stimuli have to be processed[57]. When more complex networks have to be accessed, either by way of simultaneous processing or when the stimuli are perceptually more complex, such as in second-order stimuli conditions, or when more than a single attribute must be processed to make a decision, the underlying dysfunction may become more obvious and could be expressed by functional deficits[58].

In summary, the executive function and memory are the cognitive domains most affected by age[59–61]. Van Harten and colleagues (2013), in one of their longitudinal study on preclinical AD, noticed that older adults with subjective complaints manifest cognitive decline over time not only in memory but also in executive functioning. Most of the seniors often out-perform on attentional tasks that require flexible control, dividing or switching of attention among multiple inputs or tasks[62]. Indeed, older adults face greater difficulties in performing higher-level cognitive tasks that involve manipulation, reorganization, or integration of the contents of working memory. Therefore, deficits in executive functioning might explain the memory difficulties that many older adults complaint[63,64]. This is not surprising if we think that memory tasks require organization of new information, selective attention for the information that has to be encoded, the suppression of unnecessary information, and at times the maintenance and shifting of cognitive sets. Furthermore, in order to encode and retrieve new information the cognitive efficiency relies on processing speed and working memory. Evidence suggests that slow processing speed or working memory difficulties in older adults[60] impact on the accuracy of encoding new information and on the retrieval of it later on[65]. This pattern of deficits in executive tasks is associated with episodic memory decline: the most susceptible to brain damage and the most affected by normal aging[66,67]. Episodic memory depends on multiple-interacting neural networks, including the medial temporal memory complex and prefrontal cortical executive system which explains the link between memory and executive function[68,69].

In the following chapter the importance of cognitive reserve in SCD will be discussed.

The importance of cognitive reserve in older aduls with SCD

In the last decades several models for successful aging were proposed. Specifically, two distinct neurocognitive constructs have been advocated to explain the compensatory strategies employed to optimize the cognitive performance in ageing population: ‘neural reserve’ and ‘cognitive reserve’[70]. One of them, that is mostly considered in this thesis, is “cognitive reserve”. The reserve capacity represents the hardware of the brain and it is often linked to neuroanatomical measures such as the number of neurons or synapses, brain size, or the grey matter density[71]. This perspective also emphasizes discrepancy between underlying levels of age-related deterioration or pathology and observed functional and/or cognitive efficiency as one possible cause of cognitive reserve capacity[71]. The environmental factors that seems to foster “cognitive reserve” are the engagement in leisure activities, being socially active, doing constant physical activity and using a second language daily (e.g. bilingualism)[72].

Cognitive reserve attempts to account for the fact that some individuals with extensive neuropathology associated with dementia show little cognitive decline when others with similar pathology show marked deficits [70,73]. According to the model, individuals have different amounts of reserve capacity, and as the brain starts to deteriorate in later life the person with a higher reserve capacity is less affected by the deterioration. It should be noted, however, that when using pathology as a reference, little is said about reserve capacity before the onset of pathology.

The first assumption of the cognitive reserve model refers to enhanced synaptic activity or the generation of more efficient circuits of synaptic connectivity as the result of cognitive stimulation (e.g. working memory training; music therapy, bilingualism)[74]. Enrichment factors (e.g., various factors believed to be mentally stimulating) can be beneficial not only to brain plasticity but might also promote some neurotrophic factors (i.e. increase neurogenesis in the hippocampus, an area that is important for learning and memory functioning). Furthermore, it is possible that interventions such as cognitive stimulation has an impact through breakdown of amyloid plaque, which is associated with dementia progression[75]. Therefore, introducing preventive interventions or cognitive training programs, may have significant benefits for SCD population.

Intervention

A number of pharmacological treatments for improving cognitive abilities in older adults have been proposed: antioxidant, cholinesterase inhibitors [76] or anti-Amyloid-β monoclonal antibodies[77]. However, research findings on such pharmacological treatments showed minimal effects on cognitive functioning in normal aging or Alzheimer’s disease[78]. Furthermore, great uncertainty surrounds these eventual benefits taking into consideration their side effects[79,80]. On the other side, several non-pharmacological treatments, such as brain stimulation techniques for cognitively healthy older adults[81,82], for persons with MCI[83,84,85] and for subjects with mild to moderate dementia[86,87] were proposed and examined. Considerable evidence suggests that non-pharmacological cognitive treatments may be more effective than medication[88–90].

A systematic review by Smart and colleagues (2017) on nonpharmacologic interventions for SCD revealed that cognitive and behavioural interventions are a worthwhile pursuit in seniors with SCD. Furthermore, non-pharmacological treatments can result in short-term benefits in cognition (i.e. enhance their attention or working memory capacity which may improve their memory functioning too[9,5;] learn new strategies to compensate with memory difficulties[91]) and could be also tailored for subjects with SCD. Due to this proved relevance for non-pharmacological treatments in forms of cognitive intervention it would become of utmost importance to target older patients that would be more likely to succeed in such interventions first. Predicting success likelihood for training based on cognitive and behavioural treatment is then necessary and expected. There is a need to fit these cognitive based interventions with the most likely successful patients "ahead of such intervention".

Computerized cognitive training

'Cognitive training' has been defined as an intervention consisting of repeated practice on standardised cognitive exercises targeting specific cognitive domains for the purpose of stimulating cognitive function. Although cognitive training may include traditional pen and paper tasks, nowadays it more commonly takes the form of computer-based tasks, including exercises, games and virtual reality.

The use of a virtual environment such as computerized cognitive training (CCT), especially among the elderly population, is evolving as a promising new method of combating cognitive decline. CCT had increased compliance in older adults, possibly because it is easy to access, can be used directly from home, is non-invasive, relatively inexpensive and does not require particular technological skills[81]. CCT may be delivered in individual sessions or within groups, with supervision or privately at home. Various methods of computerized technology have been tested on healthy older adults and subjects with SCD, MCI or dementia[92,93].

A systematic review by Kueider and colleagues (2012) assessed the efficacy of various CCT used on healthy older adults (e.g. video games and neuropsychological software interventions) in comparison to traditional cognitive training approaches (paper-and-pencil). Video games, using and augmented virtual reality, were found to be more effective than neuropsychological software or classic cognitive training at enhancing reaction time and processing speed. These results were correlated with a large effect size. In contrast, these games were found less effective in enhancing executive functioning and memory. On the other hand, the review on neuropsychological software interventions in older adults showed great improvements with a large effect size in processing speed, verbal memory and attention. In contrast, lower benefits were noticed on working memory and visuospatial ability.

A recent meta-analysis CCT by Mingyue Hu and colleagues (2019) showed that CCT was most effective in cognitive rehabilitation, particularly in the subdomain of memory. Specific CCT was more promising than non-specific CCT (i.e. computerised cognitive activities that do not have a treatment-oriented purpose) and it was more efficacious in patients with SCD than those with MCI[94].

Hill and colleagues (2016) reviewed several studies on CCT applied to MCI population and concluded that CCT is a viable intervention for enhancing cognition[93]. From this review, it was noticed that participants of CCT groups improved significantly over the intervention period, while controls did not show any cognitive change. Most of the trials (70%) used an active control condition and the effects across active- and passive-controlled trials were comparable. The results of the MCI analysis are therefore robust and indicate a beneficial therapeutic role for CCT in this population[93].

Another review, examining the effects of CCT compared to active or inactive controls on cognitive function in adults with MCI, has produced contradictory results. Eight randomised controlled trials (total of 660 participants) were reviewed and CCT appeared to improve performance on global cognition, episodic memory and working memory compared to active controls. However, these results are based on very low-quality evidence. No evidence was found for effects on the cognitive subdomains of speed of processing, verbal fluency and executive function, nor on functional performance, quality of life, depression, and serious adverse events, although, again, a high level of uncertainty is associated with all these results.

In summary, the main benefits of CCT were improvements in memory[95,96], working memory, processing speed and attention[97–100]. Indeed, CCT was found to be as effective as the traditional cognitive training, but less labour-intensive alternative. Furthermore, CCT had increased compliance in older adults, possibly because it is easy to access, can be used directly from home, is non-invasive, relatively inexpensive and does not require particular technological skills[81].

Virtual reality used in cognitive rehabilitation

Recently, a new cognitive rehabilitation technique was suggested and investigated in neuropsychology: virtual reality (VR). VR consists of a set of informatics technologies that create interactive realistic environments that makes the rehabilitation program an enjoyable experience. Together with telemedicine, robotics, and computer-based rehabilitation VR constitutes a new frontier of rehabilitation[101], and presents potential advantages to the rehabilitation team. VR adapts to the patient's difficulty and controls his/her performance with visual and auditory feedbacks. Moreover, these systems potentiate the quality of rehabilitation sessions, as they provide the opportunity to propose playful activities, thus increasing motivation and involvement[102].

VR systems and applications seek to address various diverse treatments for patients with MCI and dementia, such as navigation and orientation[103], cognitive functioning[104,105] as well as other instrumental activities of daily living[106,107]. Some VR systems and applications for patients with MCI and dementia have been developed in recent years[106]. A systematic review by Garcia-Betances and colleagues (2015) suggests interesting benefits of VR on attention, executive function, memory[108] and spatial orientation[109]. However, there are insufficient evidence of the benefits of use of VR in patients with MCI and further research is required[106].

Perceptual Cognitive Training

Perceptual-cognitive training (PCT), so called NeuroTracker, is an individual cognitive and perceptual training that relies on VR system. PCT is the result of more than 20 years of research by Professor Jocelyn Faubert at the

Université de Montréal. Faubert incorporated various technologies in PCT: eye- and motion-tracking systems, physiological and brain activity monitors, driving simulators and VR devices. This VR system trains the subject's ability to track and focus on multiple moving objects in the three-dimensional visual field and it was proposed to improve cognitive performance in subjects with neurological dconditions[110–113].

Several studies on PCT suggest benefits in memory, processing speed, working memory and cognitive flexibility were found in previous studies on PCT intervention[110,114–116] in different populations (e.g. healthy young adults and students with neurodevelopmental conditions, healthy adults and adults with concussions, healthy older adults and older adults with subjective memory complaints). For example, a case study on an 80-year-old man with memory complaints, that underwent 32 sessions of training with PCT, showed improvements in working memory, episodic memory, processing speed, as well as reduction in cognitive complaints with positive impact on quality of life[117]. Other work from our laboratory on healthy older adults indicated improvements in cognitive flexibility after just 7 sessions (i.e. 21 trials) of PCT[118]. Parsons and colleagues (2016) found that students who performed 10 sessions of PCT improved in performance as investigated with standardized cognitive assessments of working memory and attention on visual information. Tullo and colleagues (2018) observed that performing 15 sessions of PCT was associated with increased attentional abilities in students with neurodevelopmental conditions (e.g. Autism Spectrum Disorder, Attention-Deficit/Hyperactivity Disorder, Intellectual Disability, Specific Learning Disorder. Similarly, etc.). Vartanian and colleagues (2016) trained military personal with the PCT and observed improved performance on working memory task compared to no improvements from participants who underwent PCT training.

In summary, PCT engages visual scanning, focal attention, dynamic attention, processing speed, working memory, inhibition ability, and cognitive flexibility. All these executive processes are involved in higher-level cognitive tasks[119] as well as in memory functioning[51,60]. Because memory decline in older adults has been linked to deficits in executive processes (e.g. attention, inhibitory functions, cognitive flexibility, working memory), it was postulated that PCT may be a helpful training regimen for older adults with SCD[119]. Furthermore, predicting success likelihood for a cognitive intervention such as PCT is then necessary and expected considering all cognitive intervention proposed in the last decade. Today, there is a need to fit these cognitive-based interventions with the most likely successful patients “ahead of such intervention”. Such predictive assessments are calling upon artificial intelligence (AI) methods and machine learning techniques. The following section discusses about these approaches.

Artificial intelligence in healthcare

AI aims to mimic human cognitive functions. AI is bringing a paradigm shift to healthcare, powered by increasing availability of healthcare data and rapid progress of analytics techniques. It can be applied to various types of healthcare data: structured and unstructured. Popular AI techniques include machine learning methods for structured data, such as the classical support vector machine and neural network, and the modern deep learning, as well as natural language processing for unstructured data[120].

Medical diagnostic reasoning is a very important application area of AI[121,122]. In this framework, expert systems and model-based schemes provide mechanisms for the generation of hypotheses from patient data. For example, rules are extracted from the knowledge of experts to construct expert systems. Unfortunately, in many cases, experts may not know, or may not be able to formulate, what knowledge they actually use in solving their problems[123]. Symbolic learning techniques are used to add learning, and knowledge management capabilities to expert systems: given a set of clinical cases that act as examples, learning in intelligent systems can be achieved using Machine Learning methods that are able to produce a systematic description of those clinical features that uniquely characterize the clinical conditions[124].

Machine Learning is a technique for recognizing patterns that can be applied to medical information (e.g. patient's medical history, background information, imaging data, etc.). Machine learning typically begins with the machine learning algorithm system computing the image features that are believed to be of importance in making the prediction or diagnosis of interest. The machine learning algorithm system then identifies the best combination of these image features for classifying the image or computing some metric for the given image region. There are several methods that can be used, each with different strengths and weaknesses. There are open-source versions of most of these machine learning methods that make them easy to try and apply to images. Several metrics for measuring the performance of an algorithm exist; however, one must be aware of the possible associated pitfalls that can result in misleading metrics[123,125].

Major disease areas that use AI tools include cancer, neurology and cardiology[120]. For example, the AI application in stroke covers three major areas: early detection and diagnosis, treatment and outcome prediction, and prognosis evaluation. A review by Magoulas and Prentza (2001) reported how Machine Learning was applied for the interpretation of imperfect continuous data, used in the Intensive Care Unit to create an intelligent alarming when high risk of disease is identified. This kind of prediction resulted in effective and efficient monitoring[126]. Irimia and colleagues (2012) reviewed several studies on AI application on neuroimaging data that developed predictive methods that can be used to identify biomarkers of TBI outcome. Qiu and colleagues proposed an optimal big data sharing algorithm to identify high-risk patients which can be utilized to reduce medical cost since high-risk patients often require expensive healthcare. Another study proposed a very innovative system of prediction-based healthcare applications, including health risk assessment and dignosis[127].

Personalized therapy and outcome prediction

With big data growth in biomedical and healthcare communities, accurate analysis of medical data benefits early disease detection and patient care. Machine Learning is an effective tool in combining individual health data with predictive analytics by developing personalized treatments[124]. Currently, physician may struggle in choosing a specific treatment between the several treatments proposed. Similarly, the physician may be limited in estimating the efficacy of a specific treatment on the patient. With this purpose, Machine Learning is making great strides in medicine by leveraging patient medical history and background information to help generate multiple treatment options[123,126]. In the coming years, we will see more devices and biosensors with sophisticated health measurement capabilities hit the market, allowing more data to become readily available for Machine Learning-based healthcare technologies.

In summary, predictive analytics provides significant operational level advantages for healthcare providers. The ability to quickly and accurately diagnosing and personalizing patient treatment the first time they are admitted, boosts the patient confidence in the healthcare system, fosters the patient-doctor relationship and reduces the incidence of costly readmissions[125]. Therefore, Machine Learning can be an useful tool that can help personalizing the treatment, predict the treatment's outcome and the disease prognosis of patients with neurodegenerative diseases[123,125,128] or subjects with sub-clinical conditions (e.g. SCD). An example can be the consideration of the background factors that contribute to the amount of the "cognitive reserve" in the prediction of the efficacy of a specific cognitive treatment. Being educated, having higher levels of social interaction, working in cognitively demanding occupations, doing constant physical activity or others leisure activities, using more than one language in the daily life, are sources of cognitive reserve[129–131]. All these sources are positive modulators of neural plasticity which delays the onset of normal cognitive decline and dementia[132]. In contrast, there are background features, such as concussion or stroke events encountered over the life, that have been negatively associated with cognitive functioning[133–137]. A current machine learning research by Baranzini suggests that a patient's risk pattern (e.g. stroke, concussion history, high blood pressure) and patient's background features (e.g. age, education, bilingualism, social participation, engagement in leisure activities, etc.) might be modelled in machine learning terms as the combination of patient's multiple features that reliably predict a target outcome[138]. And this model is here addressed in order to "personalize" the cognitive intervention response on subjects with SCD.

AI models in this dissertation thesis are centred around training multiple Machine Learning algorithms with two main research strategies:

1. Identify if there exists a subset of variable/feature predictors like patient background data in order to make reliable predictions on cognitive skills or on PCT training performance indicators;
2. Derive insights (interpretation) about the importance and direction of such sets of predictor features (i.e., the role and impact of each predictor in model). For example, we expect to obtain predictive models to facilitate the decision-making capacity of the neuropsychologist/physician to weather expose or reject a patient to the PCT intervention.

In conclusion, the core concept is that these algorithmic models are not only successfully developed and trained, but more importantly from a clinical point of view, such models can then be deployed in real diagnostic settings at work. Machine Learning trained parameters or decision rules will then be used to know in advance what will be the most likely PCT or cognitive performance of an older adult with SCD with certain background data configuration of values (i.e. an useful pre-screening to help with decision making on whether to engage a patient PCT intervention or not). Furthermore, very relevant is that the neuropsychologist would be driven by the Machine Learning algorithms to focus only on those subsets of all potential variables most likely impacting the elderly patients. Thus, this thesis would apply AI methods to detect if such operating models exist (i.e. by the identification of the most important features in the total set of patient data, environmental factors, and medical history data available) and may predict the efficacy of Perceptual-Cognitive Training in older adults with SCD.

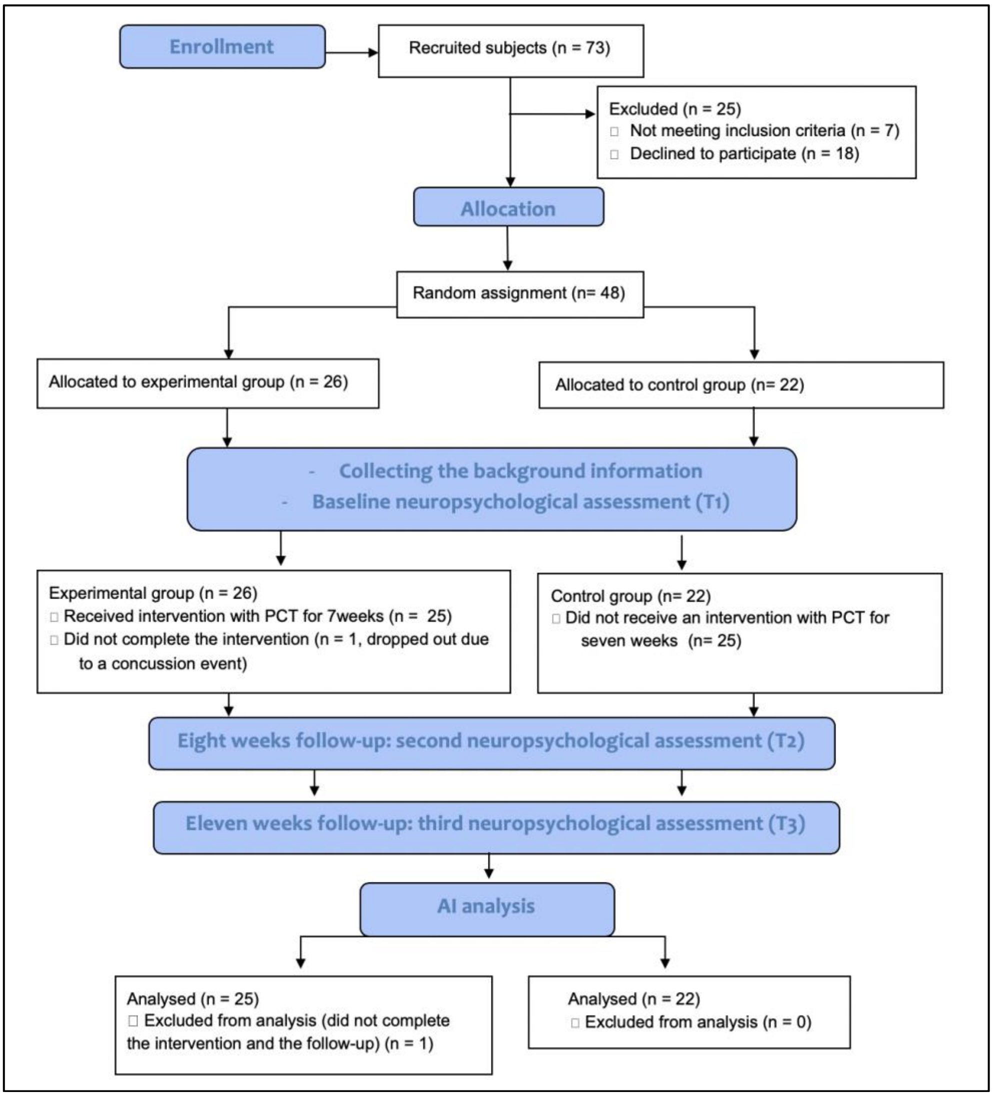
Methods and Materials

Participants

A sample of 73 participants, between 60 and 90 years of age, was recruited using word of mouth referral and flyer distribution in the Capital Regional District (CRD) encompassing the southern tip of Vancouver Island. A print and web-based advertising were also used through the Institute on Aging and Lifelong Health at the University of Victoria.

Socio-demographic information was collected from all participants at the baseline session (e.g., age, gender, level of education, and medical history) by completing an intake form approved by ethics committee of University of Victoria. All participants were screened for any medical, neurological, or psychiatric conditions known to affect cognitive performance in the first interview. The Mini Mental State Examination[139] was used as a screening tool (cut-off ≥ 26) to minimize the risk of including persons with preclinical dementia but as well to quantify the subjective cognitive complaints. Two tests, Activities of Daily Living[140] and Instrumental Activities of Daily Living [141], were administered to exclude subjects with possible dementia and to ensure that they were able to attend the testing and the training sessions at the University of Victoria. All participants were screened for SMCs using the Memory Complaint Questionnaire[142] and only the participants with a score of 25 or above were included in this study (see *Appendix A*). All participants were screened for depression using the Geriatric Depression Scale[143] with cut-off ≥ 10, and for anxiety using the Geriatric Anxiety Inventory[144] with cut-off > 9. On self-report of a diagnosis, seven participants did not meet the inclusion criteria (i.e. one had ADHD, four subjects had Macular Degeneration; one had Anxiety Disorder, one had PTSD) and were not included in this research study. Following participant screening, only 66 subjects (female n = 48, 72.2%), aged 60 years and over ($Mean_{Age}$ = 73.32, SD_{Age} = 7.58) satisfied the inclusion criteria and were enrolled in the study (e.g. over a three-month period). Eighteen subjects declined their participation to the study due to a long commitment time required. One female participant dropped out during the study due to a neurological event (e.g. a concussion outside the testing environment) and her data was removed from the analysis. The remaining 47 subjects (see Figure 3) were randomly assigned to either the treatment or control group and all subjects completed the follow-up. The method used to generate the allocation sequence was self-selection (i.e. we generated a random assignment based on the participant's availability to commit to the study). The treatment group consisted of 25 participants between the ages of 61 and 89 years of age (female = 16; male = 9) whereas the control group consisted of 22 older adults, ages 60 to 90 years (female = 15; male = 7).

Figure 3. Flow diagram summarizing patient recruitment and progress of the study.



Procedure

This clinical study, using a parallel design, was approved on 27th June 2017 by the University of Victoria Human Research Ethics Board. This study was not initially registered as a clinical trial as the UVic Research Ethics Board did not consider this study as a clinical trial but as a research study on sub-clinical population (e.g. Subjective cognitive decline). In response to a request from a journal editor, the authors registered this intervention as a clinical trial at clinicaltrials.gov (NCT03763344).

All participants provided their informed written consent prior to participating in this study. Participants from both the treatment and control groups received a total of three neuropsychological assessments over a three-month period (see Figure 3). All the data were collected in the Concussion Laboratory of the Division of Medical Sciences, at the University of Victoria. All the tests were administered by a Doctoral Student in Clinical Neuropsychology, as this was considered an essential methodological component of the training studies[145], and standardized neuropsychological tests were used to provide validated and reproducible measures. The primary outcome measure was California Verbal Learning Test, Second Edition (i.e., standard and alternate forms)[146–148]. The secondary outcome measures were Digit Span, D-KEFS Trail Making Test, D-KEFS Verbal Fluency Test (both standard and alternate forms)[146–148], and Stroop Test. Each assessment was 50-60 minutes in duration and was administered by a Doctoral Student in Clinical Neuropsychology. The first assessment was administered at baseline (T1). Then, each subject of the treatment group underwent seven weeks of perceptual cognitive training, while the control group completed seven weeks without formal training. The intervention consisted of 14 sessions of PCT each lasting 25-30 min, twice per week for seven weeks. After the seven-week time period, a second neuropsychological assessment was performed on both groups (T2). After eleven weeks, a follow-up assessment was conducted to verify whether the benefits of cognitive training endure over time (T3).

The results observed at T2 and T3 were used to make an accurate personalized outcome prediction (e.g. Machine Learning models) to facilitate the decision-making capacity to weather expose or reject a patient to the PCT intervention. For these Machine Learning applications it was decided to rely on previous research data collected (e.g background information) and cognitive results obtained in the author previous research (Musteata et al., 2019). Particular attention was dedicated to study the effects of subject's background data (e.g. age, education, concussion history, stroke history, bilingualism, leisure activities, etc.) and specific cognitive tests as potential predictors of these new Machine Learning models (see the objectives 1, 2, 3 in the Data Analysis chapter). In fact, the impact of the individual characteristics is less known in the prediction of cognitive performance (e.g. CVLT-II Immediate Recall Scores, CVLT-II Long Delay Recall Scores, etc.) and more importantly, in the predictive strength of the actual training intervention outcome (e.g. PCT).

Materials

Questionnaires

The **Memory Complaint Questionnaire (MAC-Q)** is a self-report tool, developed in the hope that an individual's own report on their current memory ability relative to the past might provide a clue as to the stability or otherwise of memory performance over time (see *Appendix A*). The MAC-Q was developed with the intent of screening for Age-Associated Memory Impairment (AAMI) and for research that requires an index of memory complaint. The MAC-Q consists of six items. The first five items relate to specific situations that are frequently reported as troublesome for those with declining memory, and the last item broadly measures overall self-perceived memory decline. A score of 25 and above on the MAC-Q is indicative of significant memory complaint that may require further assessment[149].

The **Mini-Mental State Examination (MMSE)** is a 30-point screening test that is used extensively in clinical and research settings to measure global cognitive efficiency (see *Appendix B*). The items of the MMSE include tests of orientation, registration, recall, calculation and attention, naming, repetition, comprehension, reading, writing and drawing. Any score of 24 or more (out of 30) indicates a normal cognition. Below this, scores can indicate severe (≤9 points), moderate (10–18 points) or mild (19–23 points) cognitive impairment.

The **Leisure Activity Questionnaire** investigates the participation in 16 leisure-time activities during the last 3 months: Travel/Trips; Sports/Exercise/Walking in the forest; Hunting/Fishing; Read books; Read magazines; Read newspaper; Watching TV/Listen to the radio; Movies/Concerts/Theatre; Restaurant visits; Spend time with family, relatives, and friends; Attending Courses/Workshops; Religious assemblies; Association work; Play musical instrument; Needlework; and Other Hobbies/Activities (see *Appendix C*). For each activity is indicated the frequency of participation on an ordinal scale "never" (coded as 0), "occasionally" (1) , "a few times a month" (2), "sometime per week" (3), and "every day" (4).

The **International Physical Activity Questionnaires for Elderly (IPAQ-E)** is a short version of the original self-administered IPAQ (see *Appendix D*). The purpose of the questionnaire is to investigate four types of physical activity: vigorous activity, moderate activity, walking activity and an indicator of sedentary behaviour (e.g. time spent sitting). Activities such as housework, gardening, leisure-time, sport and exercise. Moderate intensity was defined as 3–6 MET (Metabolic Equivalent Task) and vigorous intensity was defined as .6 MET12. One MET is equal to energy expenditure during rest and is approximately equal to 3.5 ml O2 kg21 min21 in adults.

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Neuropsychological Tests

The **California Verbal Learning Test Second Edition**[150] (CVLT-II) is a multiple-trial list-learning task that measures individuals' *episodic memory* and *auditory learning ability*. CVLT is considered a sensitive tool in identifying subtle episodic memory difficulties. This test assesses recall and recognition of two-word lists over immediate and delayed memory trials. Standard and alternate forms of these lists exist, each with different lists of words to avoid practice effects. Each form contains two lists: list A and list B. List A is composed of 16 words divided in four different semantic categories (e.g., furniture, vegetables, methods of transportation, and animals); whereas words from the same semantic category are never presented consecutively. There are five trials using List A, and each trial requires the participant to immediately recall as many words from the list as possible. List B is a 16-word interference list, which includes different categories. List B is presented once, following the five trials of immediate recall of List A. Immediately after presentation of List B, short-delay free recall of List A is administered. Between the short-delay recall and long-delay recall, there is a 20-minute delay, which is filled with non-verbal testing (e.g., D-KEFS TMT; Stroop Test). After the non-verbal testing, long-delay free recall of List A and a recognition task (yes/no format) are administered. This list included words from both List A and List B, as well as other distractor words, where the examinee is required to identify only the words belonging to List A.

The **Delis–Kaplan Executive Function System Trail Making Test**[151] (D-KEFS TMT) is a pencil and paper task, used to evaluate aspects of cognition including *processing speed, motor speed* and *cognitive flexibility*. It involves a series of five conditions: visual scanning, number sequencing, letter sequencing, number-letter switching, and motor speed. In the visual scanning condition, examinees must cross out all the threes that appear on the response sheet mixed with other numbers. In the number sequencing condition, examinees draw a line connecting the numbers 1–16 in counting order while avoiding distractor letters that appear on the same page. The letter sequencing condition requires examinees to connect the letters A through P, with distractor numbers presented on the page. In the number-letter switching condition, examinees switch back and forth between connecting numbers in counting order and letters in alphabetical order (i.e., 1, A, 2, B, etc., to 16, P). This condition requires the ability to switch mentally between numerical and alphabetical sequences and provides an assessment of the participant's cognitive flexibility. Finally, the examinee completes a motor speed condition in which he/she has to trace over a dotted line connecting circles on the page as quickly as possible. This final section assesses their graphomotor speed.

The **Delis–Kaplan Executive Function System Verbal Fluency Test** (D-KEFS VFT)[151] is a short test of verbal functioning that measures *processing speed* and *cognitive flexibility*. There are three conditions: Letter Fluency, Category Fluency, and Category Switching. In all three conditions, the examinees are given 60 seconds to generate as many words following a semantic cue (e.g., specific category), a phonemic cue (e.g. starting with a certain letter) or alternating between two categories a task, which requires a certain amount of mental flexibility.

The **Digit Span Test**[152] is a measure of *working memory* consisting of 16 trials; eight in Digit Span Forward, and eight in Digit Span Backward. In both conditions, the examiner reads out a series of numbers, ranging from 2-9 digits in sequence. In the forward condition, the participant is asked to repeat the numbers verbatim as stated by the examiner at the end of each trial. In the backward condition, the participant is asked to repeat the numbers in the reverse order stated by the examiner.

The **Stroop Test** is used to measure *selective attention*, *psychomotor speed* and *cognitive flexibility* [153,154]. In this study, the Stroop test was delivered using the Encephal App, which adheres to the same principles as the classic Stroop version[155]. Subjects are required to identify the ink colour of discordant-colour words (red, blue, or green). The task consists of two parts: the Stroop effect turned off (i.e. the examinees name the colour of the ink of a set of number signs) and the Stroop effect turned on (i.e. series of colour words “Red”, “Blue”, “Green” are presented in an incongruent coloured ink). In this task, the examinee must inhibit the automatic tendency of reading in order to name correctly the colour of the ink. The placement of the words and number signs are randomized and change position on the screen with each new stimulus. The order of the responses on the bottom of the screen that examinees need to respond to are randomized and shifts in order with each new stimulus. The examinees are not instructed that the order of the response options shift with each new screen, requiring more focus and mental flexibility to the changing stimuli.

Intervention: Perceptual-Cognitive Training

Perceptual-cognitive training is a virtual reality system developed by Jocelyn Faubert of University of Montreal[56,114,156]. This training is based on a computerized 3D Multiple Object Tracking (3D-MOT) model that follows two principles: isolation and overloading. Isolation training uses limited and consistent cognitive load, while overloading challenges the subject by training them at levels beyond their current ability in order to increase cognitive functioning. Previous studies have indicated that the training effect is reduced if isolation and overloading are not applied to the task[157,158].

Each PCT session consists of three series of 20 trials in which the subject wears 3D glasses and tracks four spheres among four identical distractors that move in a 3D volumetric cube on the screen. In the first phase, all eight spheres are stationary on the screen, then the four targets briefly change to red and after two seconds revert to yellow. The four target spheres must be tracked as they move in a linear trajectory for eight seconds. After this, the spheres stop moving and the subject is asked to identify the four targets.

The sessions are based on a staircase procedure [159], in which an algorithm shifts the speed of the target spheres in regard to the participants' performance (i.e., overloading principle). If all targets were correctly identified, the speed of the movement of the spheres increases by 0.05log, whereas with each incorrect response the speed decreases by 0.05log.

Data Analysis

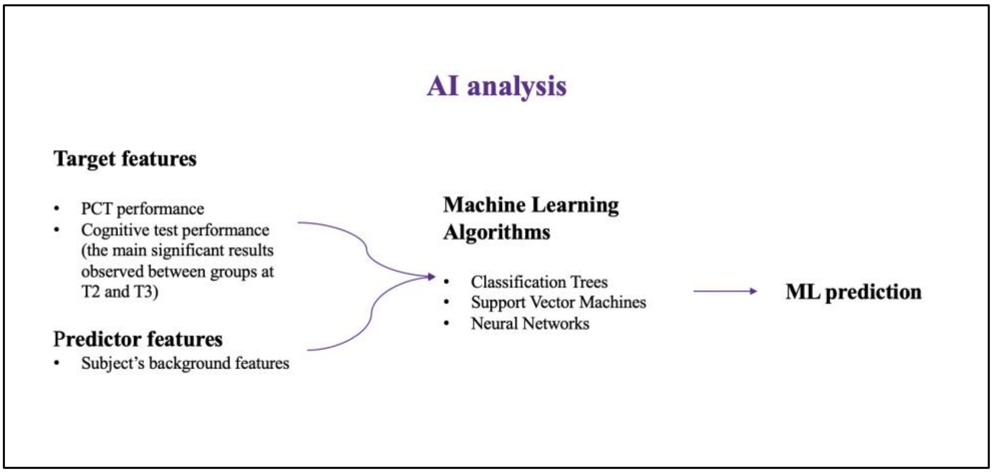
IBM SPSS Statistics v25.0 and IBM SPSS modeler 18.1 were used for the statistical and machine learning analyses. Standard descriptive statistics and data screening were computed. Full statistical diagnostics and data cleaning were carried out to check for adequate distributions, adjustments of out-of-range scores, missing values, outliers as well as appropriate standard deviations and proper standard errors values. Such diagnostics were iteratively conducted on the background data, cognitive performance data (e.g. prior the PCT intervention (T1), after the seven weeks of training (T2)) and on the training performance data. In particular, box plots for each group and dependent measure were used to identify critical outliers pre-, post-training, and after a month of follow-up. The Trimming method[160,161] was used to constrain response values with more than 3 standard deviations above or below the mean. Data of a subject that drop out in the middle of the intervention for a concussion reason was removed.

The expectations in this dissertation thesis are the following:

1. to make AI predictive models learn an accurate combination of specific predictors (i.e. a pattern of background variables) and thus anticipate (predict) patient's scores on targeted cognitive tests (and relative accuracy);
2. to make AI predictive models detect a set of predictors (i.e., background and cognitive tests features) and thus make predictions on subject's PCT performance levels (and relative accuracy);
3. to finally deploy algorithms "trained" in 1) and 2) on new incoming patients and thus anticipate (in advance of the cognitive training) the expected scores on cognitive tests and PCT performances.

These AI analysis (see Figure 4) were based on some of the collected data in the study of Musteata and colleagues (2019). Specifically, the performance of the experimental group obtained in the PCT and in the cognitive tests was used. Note, the predictive models were developed on only 25 cases, which remains a too small sample to consider cross validation tests sensibility (i.e. the generalization does not occur directly by the applications). Nevertheless, such predictive modelling outcome is proposed in order to provide evidence of the potential and leverage that AI oriented methods can directly bring into so called “predictive modelling and Machine Learning sensitivity analysis for personalized intervention in neuropsychology”.

Figure 4. AI analysis



Results

Predictive treatment outcome by Machine Learning application

Cognitive Test Score Prediction

A large Machine Learning model search was carried out to maximise prediction scores about each target cognitive test (i.e. ratio of the variance of the observed values from those predicted by the model to the variance of the observed values from the mean presented in previous research Musteata et al., 2019).

The target variables scores to make predictions are (see *Table 4* and *5* in the manuscript Musteata et al., 2019):

- *CVLT-II Immediate Recall,*
- *CVLT-II Short Delay Recall,*
- *CVLT-II Long Delay Recall,*
- *D-KEFS Verbal Fluency Test: Letter Fluency,*
- *D-KEFS Verbal Fluency Test: Category Switching.*

The potential predictors are (see Table 1 and Table 2):

- *Age,*
- *Gender,*
- *Years of Education,*
- *Bilingualism,*
- *Concussion History,*
- *Stroke History,*
- *MMSE,*
- *MAC-Q,*
- *Leisure Activities: Total Index,*
- *Leisure Activities: Social Index,*
- *Leisure Activities: Physical Activity Index,*
- *Leisure Activities: Mental Index,*
- *Leisure Activities: Other Activities Index,*
- *IPAQ: Total Score,*
- *IPAQ: Vigorous Score,*
- *IPAQ: Moderate Score,*
- *IPAQ: Walking Score.*

Table 1. Demographic information of background variables used as predictors

Treatment group (n=25)					
	AGE	GENDER	YEARS OF EDUCATION	MMSE	MACQ
MEAN	74,75	16(F)/25	16,4	29,24	27,6
SD	8,73	-	4,03	1,30	2,38

Table 2. Demographic information of background variables used as predictors

Treatment group (n=25)													
	BILINGUALISM	CONCUSSION HISTORY	STOKE HISTORY	LEISURE ACTIVITIES: TOTAL INDEX	LEISURE ACTIVITIES : SOCIAL INDEX	LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX	LEISURE ACTIVITIES: MENTAL INDEX	LEISURE ACTIVITIES: OTHER ACTIVITIES	IPAQ: TOTAL SCORE	IPAQ: VIGOROUS SCORE	IPAQ: MODERATE SCORE	IPAQ: WALKING SCORE
MEAN	4/25	9/25	3/25	31,44	9,88	3,04	16,28	2,24	2947,5	734,4	1068,24	1144,88
SD	-	-	-	5,73	3,78	1,02	3,88	1,36	2206,34	1290,73	1210,27	1131,01

A set of six types of diverse Machine Learning algorithms were run and trained to predict each Target Variable for a total of five different predictive models builds. The six baseline algorithms employed were:

- CHAID Tree = Chi Square Automatic Interaction Detection Tree (a tree-based algorithm that partitions stepwise all feature space into purer branches (optimal feature differentiations) of any class number. Each algorithmic tree iteration selects the best feature to split based on its chi square statistic *p-value*[162].

- CART Tree = Classification and Regression Tree (a tree-based algorithm that partitions the measurement space only into binary classes at each split node. The stepwise branching maximises information entropy reduction (e.g., Cross-Entropy index) at every split down the tree generation[163].

- SVM = Support Vector Machine (a mathematical set of optimization constraining algorithms to construct an optimally separating hyperplane, a proper decision boundary, for target classes in the measurement space)[164].

- LSVM = Linear Support Vector Machine (a mathematical set of optimizations constraining algorithms to construct an optimally separating hyperplane, a proper decision boundary, for regression predictions problems in the measurement space)[164].

- NN = NEURAL NETWORK (a formal stage-wise system of input-hidden-output layers of distrusted chained regression or classification algorithms which iteratively adjust parameters weights by retrofitting information about the error rate in prediction or classification[164,165].

- *k*-NN = *k*-Nearest Neighbour (a *memory-based* classification technique that iteratively searches for k data instances with the closest distance from a target point of reference and then classify the k neighbours by majority vote)[165].

For each of the five individual target features, all six machine learning algorithms were run to compete and be ranked by their relative error rates in prediction and their correlation value (*Pearson r*) between the observed and predicted scores. The best three ranked models (per target variable) were selected and reported in Table 3, Table 5, Table 7, Table 8, Table 9 respectively. To note that the first two of the original six baseline algorithms above are tree classifiers. Such tree-based algorithms are designed to be "readable models", that is, the combination and effects for each predictor internal to the model is explicit and interpretable: it is always possible to explain the internal model structure or effects across relevant predictors. In contrast, Machine Learning models like SVM, LSVM and NN or k-NN are called sometimes "black-boxes" as their internal model behaviour is not explicit" or directly "readable". Individual role and impact of each predictors is not directly interpretable.

Following on from this rationale, tree algorithms like CHAID were primarily used in this thesis to interpret impact and role of predictors (see Table 3, Table 5, Table 7, Table 8, Table 9) despite the actual ranking of models' performance by correlation and error rates criteria (see Figure 5, Figure 7, Figure 9). The importance of the target predictors has been evaluated with 2 different methods:

1. One was to consider the best five subject's predictors where the importance of each one of the five predictors was assessed in combination with the others (i.e. this is a relative importance criteria of prediction) (see Figure 6, Figure 8, Figure 10)

2. The decision trees provides prediction importance based on each variable taken independently (see Table 4, Table 6, Table 10).

Target: CVLT-II Immediate Recall

In Table 3 the best model prediction for CVLT-II Immediate Recall is for CHAID Tree classifier with a correlation value of 0.895 and relative error at 0.199. Figure 5 shows the CHAID scatterplot between the observed and predicted values correlation and a measure about the most important predictors (top five) to build this tree model: IPAQ: Vigorous Score, Age, MACQ, MMSE and IPAQ: Walking Score are very important features for prediction of this cognitive test score (Figure 6).

Table 3. Target CVLT-II Immediate Recall

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	CHAID 1	< 1	0.895	5	0.199
2	SVM 1	<1	0.647	17	0.655
3	KNN Algorithm 1	< 1	0.528	17	0.741

Figure 5. CHAID scatterplot between the observed and predicted values correlation and a measure

about the most important predictors

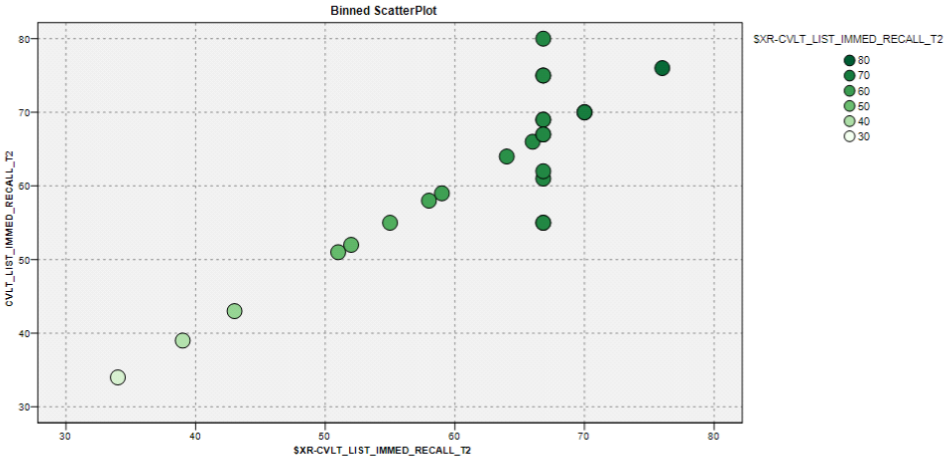
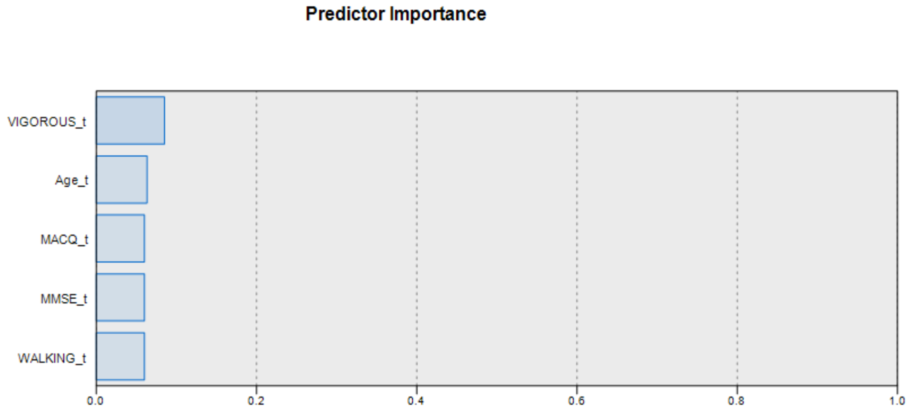


Figure 6. Top five predictors CVLT-II Immediate Recall



Predictor Importance = normalized relative importance from 0 to 1 value

More interestingly, the tree classifier partitions the measurement space into simple decision rules, advanced patient's profile patterns indicating what kind of patient profile is predicting higher or lower CVLT-II Immediate Recall score. An example of the total decision rules is shown in Table 4.

For instance, if a patient is Female with score above 30 on Leisure

Activities Total Index measure then his/her predicted CVLT-II Immediate Recall is of 66.82 points. Instead if the patient is a Male, with no Vigorous values (< = 0) and with Leisure Activities Total Index ranging between 30 and 31 then then his/her predicted CVLT-II Immediate Recall is remarkably lower at 39 points.

Table 4. Decision rules

DECISION RULE 1	
IF	Gender = Female
AND	LEISURE ACTIVITIES TOTAL INDEX t > 30
THEN	CVLT-II Immediate Recall= 66.82 (44% of sample)
DECISION RULE 2	
IF	Gender = Male
AND	IPAQ VIGOROUS SCORE t < = 0
AND	LEISURE ACTIVITIES TOTAL INDEX t > 30 and LEISURE ACTIVITIES TOTAL INDEX t < = 31
THEN	CVLT-II Immediate Recall = 39.00 (4% of sample)

Target: CVLT-II Short Delay Recall

In Table 5 the best model prediction for CVLT Short Delay Recall is for a Neural Network with a correlation value of 0.885 and relative error at 0.220. As a Tree classifier, CHAID Tree, is ranked second within the top 3 predictive models, the interpretation is based on such a readable model and not the best model Neural Network, a "black-box" system for feature interpretation. Thus, Figure 7 shows the CHAID scatterplot between the observed and predicted values correlation and a measure about the most important predictors (top five) to build this tree model: MACQ, MMSE, IPAQ Walking Scores, IPAQ Vigorous Scores, and IPAQ Total Scores. are very important features for prediction of this cognitive test score.

Table 5. Target CVLT-II Short Delay Recall

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	Neural Net 1	< 1	0.885	17	0.22
2	CHAID 1	<1	0.727	3	0.471
3	SVM 1	< 1	0.747	17	0.578

Figure 7. CHAID scatterplot between the observed and predicted values correlation and a measure about the most important predictors

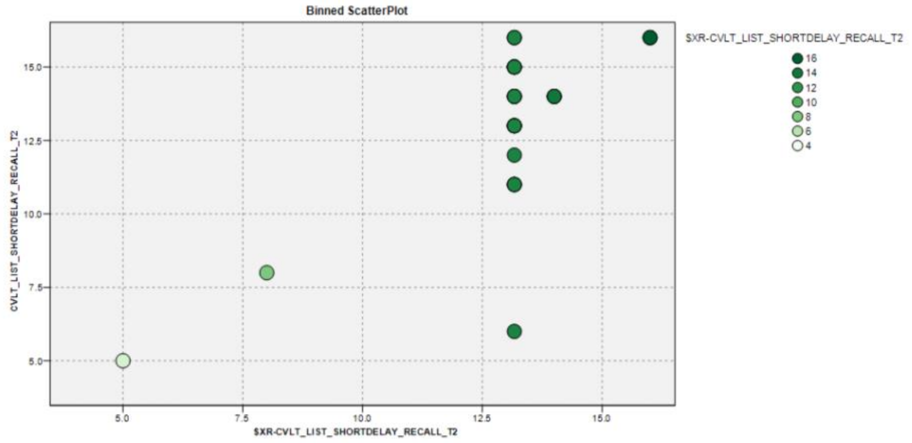
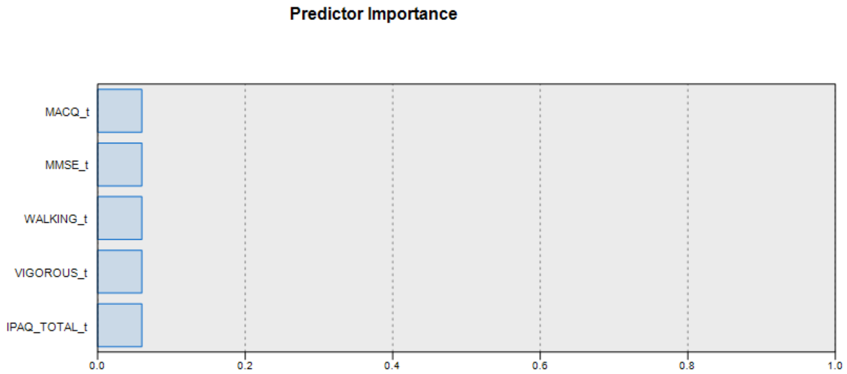


Figure 8. Top five predictors for CVLT-II Short Delay Recall



Predictor Importance = normalized relative importance from 0 to 1 value

The tree classifier decision rules in Table 6 indicates that if a patient has IPAQ: Moderate Score between 1200 and 1440 and Age is below or equal to

81 then the "predicted" score on CVLT-II Short Delay Recall is as low as 5.0. Instead if a patient has IPAQ: Moderate Score between higher than 1440 and IPAQ: Walking Score between 462 and 792 then the "predicted" score on CVLT-II Short Delay Recall is as high as 16.0.

Table 6. Decision rules

DECISION RULE 1	
IF	IPAQ MODERATE SCORES t > 1200 and IPAQ MODERATE SCORES t < = 1440
AND	Age t < = 81
THEN	CVLT-II Short Delay Recall = 5.0 (4% of sample)
DECISION RULE 2	
IF	Gender = Male
AND	IPAQ MODERATE SCORES t > 1440
AND	IPAQ WALKING SCORES t > 462 and IPAQ WALKING SCORES t < = 792
THEN	CVLT-II Short Delay Recall = 16.0 (4% of sample)

Target: CVLT- II Long Delay Recall

In Table 7, the best performance is again for a CHAID tree model. But this tree structure does overfit the data completely (correlation = 1 and relative error = 0) and, being the only "readable model" available, no predictors interpretation findings is conducted. More reliable data samples are necessary are required for this analysis. Overall, the second-best model for prediction for CVLT-II Long Delay Recall is a Support Vector Machine with a favourable correlation value of 0.839 and relative error at 0.49. Thus, reliable predictive action is possible but not model interpretation.

Table 7. Target CVLT Long Delay Recall

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	CHAID 1	< 1	1.0	6	0.0
2	SVM 1	<1	0.839	17	0.49
3	Neural Net 1	< 1	0.697	17	0.562

Target: D-KEFS VFT: Letter Fluency

In Table 8 the best model prediction for D-KEFS VFT Letter Fluency is for a Neural Network with a moderate correlation value of 0.445 and relative error at 0.826. No "readable" regression Tree (CHAID or CART) have been ranked in the top three ones. Thus, no further internal model interpretation about feature roles is given as in the previous modelling effort for the D-KEFS VFT Letter Fluency prediction.

Table 8. Target D-KEFS VFT Letter Fluency

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	Neural Net 1	< 1	0.445	17	0.826
2	SVM 1	<1	0.568	17	0.909
3	KNN Algorithm 1	< 1	0.289	17	0.942

Target: D-KEFS VFT: Category Switching

The Table 9 reveals that the best model and prediction for D-KEFS VFT Category Switching is for the CHAID Tree classifier with a correlation value of .820 and relative error below .328. The CHAID observed and predicted values scatterplot and the predictors importance for this model are in Figure 9. Leisure Activities Total Index, Gender, MAC-Q, IPAQ Walking Score and IPAQ: Moderate Score result very important features for this model construction.

Table 9. Target D-KEFS VFT Category Switching

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	CHAID 1	< 1	0.82	4	0.328
2	SVM 1	<1	0.591	17	0.699
3	Neural Net 1	< 1	0.486	17	0.785

Figure 9. CHAID scatterplot between the observed and predicted values correlation and a measure about the most important predictors

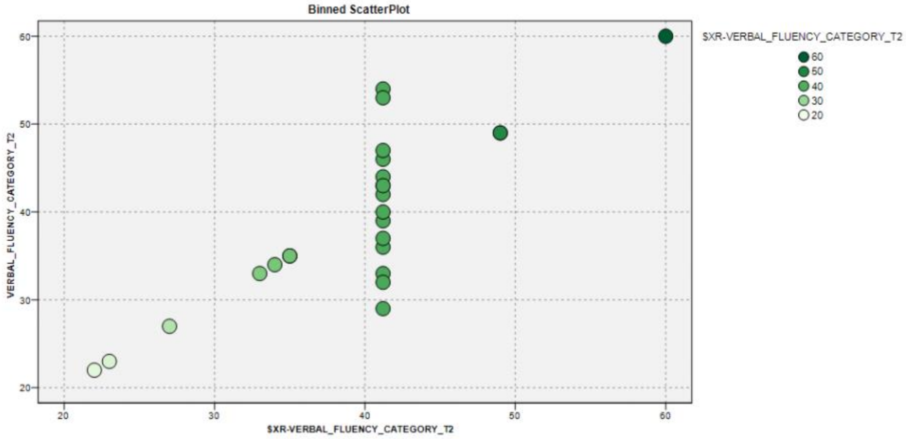
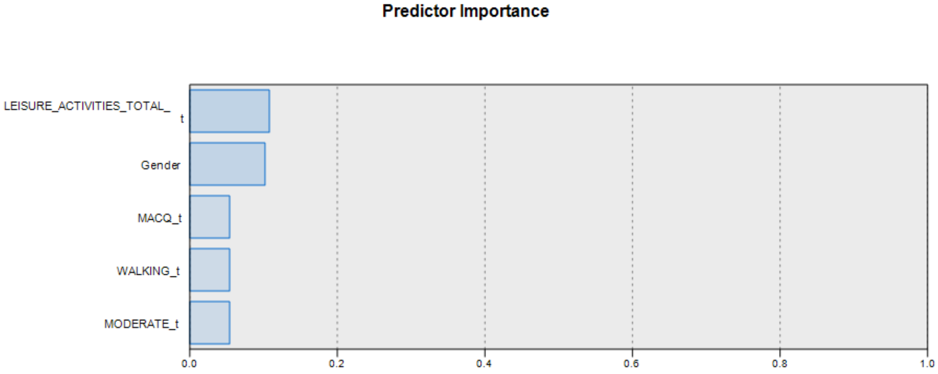


Figure 10. Top five predictors for D-KEFS VFT Category Switching



Predictor Importance = normalized relative importance from 0 to 1 value

The tree classifier decision rules in Table 10 indicates that if a patient has scores > 29 in MMSE and Gender is Male with no Concussion History then the "predicted" score on D-KEFS VFT Category Switching is of 60 points. Instead, if the patient is identical but with a Concussion History (=1), then his/her scores are predicted to drop to 49 points on the D-KEFS VFT

Category Switching.

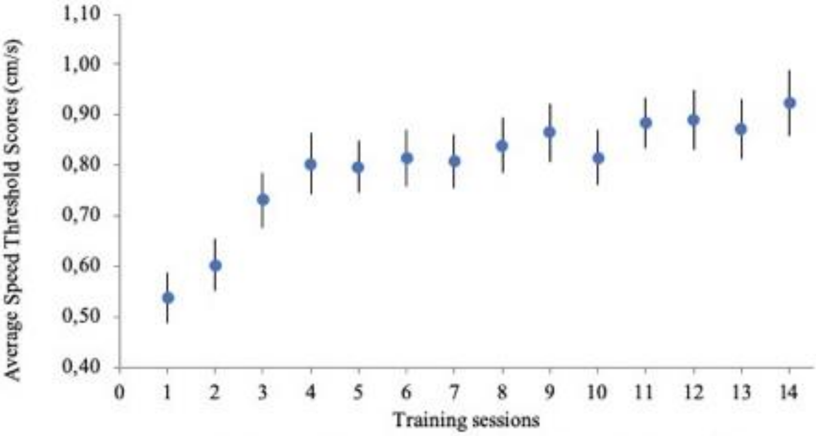
Table 10. Decision rules

DECISION RULE 1	
IF	MMSE t > 29
AND	Gender = Male
AND	Concussion history = 0
THEN	*D-KEFS VFT Category Fluency* = 60.0 (4% of sample)
DECISION RULE 2	
IF	MMSEct > 29
AND	Gender = Male
AND	Concussion history = 1
THEN	*D-KEFS VFT Category Fluency* = 49.0 *(*4% of sample*)*

Machine Learning models to anticipate the subject's pattern on PCT performance

Machine Learning models for Perceptual-Cognitive training scores (see Table 11 and Figure 11) prediction followed a similar approach as reported in the previous section. In this case, the interest in modelling prediction was driven directly by screening predictors and their "internal" relative configuration and impact on performance. Thus, the CHAID tree algorithms were directly assessed as primary AI mechanism of choice.

Figure 11. Average speed threshold scores with PCT from the treatment group



The target of this prediction is Perceptual-Cognitive Training performance as follows:

1. Training Median Performance scores: the median value of the 14 mean training sessions;
2. Training Slope Performance: the differential (gap) between the mean of the first training session and the last one;
3. Training Stability Performance: the standard deviation values of the 14 mean training sessions.

Table 11. Perceptual-Cognitive Training performance of the 14 sessions

	TRAINING SLOPE	TRAINING STABILITY	TRAINING MEDIAN
Subject 1	0,52	0,18	1,27
Subject 2	0,44	0,16	0,75
Subject 3	0,09	0,12	0,66
Subject 4	0,45	0,2	0,92
Subject 5	0,29	0,11	0,44
Subject 6	0,22	0,17	0,88
Subject 7	0,53	0,17	0,97
Subject 8	0,35	0,15	0,61
Subject 9	0,66	0,16	0,81
Subject 10	0,14	0,13	0,94
Subject 11	0,46	0,2	1,18
Subject 12	0,57	0,17	0,92
Subject 13	0,29	0,13	1,02
Subject 14	0,53	0,2	0,78
Subject 15	0,66	0,2	1,12
Subject 16	0,7	0,23	0,63
Subject 17	0,2	0,13	0,78
Subject 18	0,35	0,18	0,41
Subject 19	0,05	0,09	0,39
Subject 20	0,17	0,11	0,63
Subject 21	0,5	0,17	1,08
Subject 22	-0,13	0,11	0,61
Subject 23	0,63	0,19	0,87
Subject 24	0,12	0,1	0,43
Subject 25	0,65	0,2	1,14
Mean	0,38	0,16	0,81
SD	0,22	0,04	0,25

All these three PCT training performance criteria have been modelled as relative speed training task performances (as explained in the materials section under *"Intervention: Perceptual-Cognitive Training"*). Instead, the five previous cognitive target variables (i.e., *CVLT-II Immediate recall, CVLT-II Short Delay Recall, CVLT-II Long Delay Recall, D-KEFS VFT Letter Fluency, D-KEFS VFT Category Switching*) were now added as predictors together with all original background predictors features (Age, Gender, Years of Education, Bilingualism, Concussion history, Stroke, Leisure Activities: Total Index, Leisure Activities: Social Index, Leisure Activities: Physical Activity Index, Leisure Activities: Mental Index, Leisure Activities: Other Activities Index, IPAQ: Total Score, IPAQ: Vigorous Score, IPAQ: Moderate Score, IPAQ: Walking Score, MMSE, MAC-Q). These constitute the total set of predictors' features that has been fit to three CHAID models and the available training data set. Each trained regression tree is now reported for each model target separately.

Target: PCT Median Performance

In Table 12 the CHAID tree classifier fit to prediction of *PCT Median Performance* resulted in a correlation value of .885 and relative error of .217. The Figure 12 shows the CHAID scatterplot of observed and predicted target values and the ranked most important predictors to build this model: IPAQ Vigorous Score, Age, Years of Education, D-KEFS VFT Letter Fluency, D-KEFS VFT Category Fluency reveal to be very important features to construction of the tree model.

Table 12. Target PCT Performance

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	CHAID 1	< 1	0.885	4	0.217

Figure 12. CHAID scatterplot between the observed and predicted values correlation and a measure

about the most important predictors

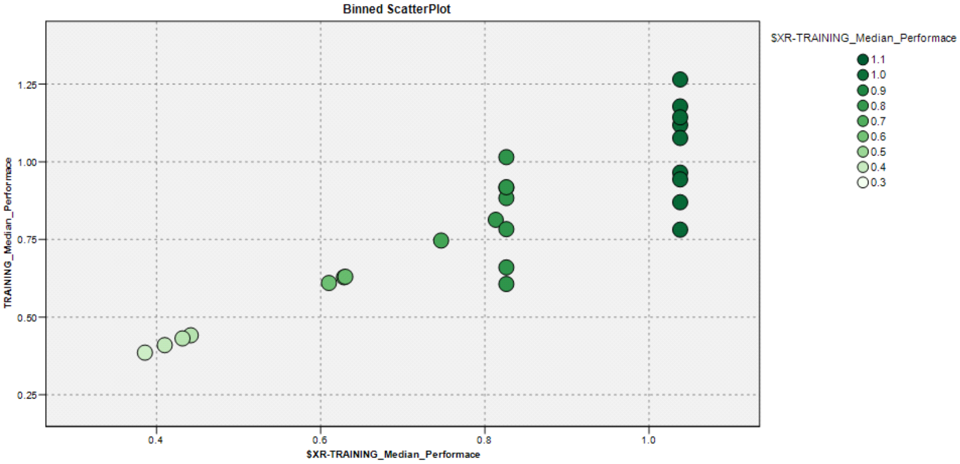
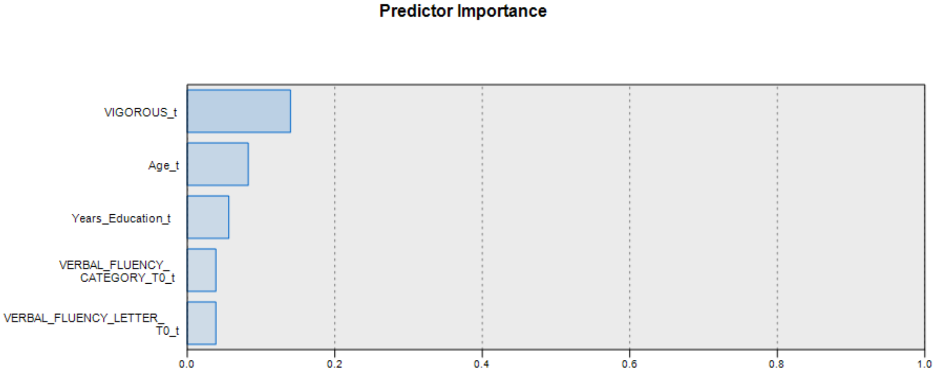


Figure 13. Top five predictors of PCT Median Performance



Predictor Importance = normalized relative importance from 0 to 1 value

Two revealing examples of the role of predictor features in this CHAID tree classifier are in Table 13. For instance, if a patient has IPAQ Vigorous Score < = 240 and Age is > 76 with Years of Education < = 11 then the predicted PCT Median Performance scores are under .386 log. Furthermore, if a similar patient for IPAQ Vigorous Score (< 240) and AGE

(> 76) shows higher scores in Leisure Activities Other Activities Index, and Years of Education its PCT Median Performance rises up to .816 log. In contrast, if such a patient simply scores on IPAQ Vigorous Score (>240) the predicted performance for training jumps to a favourable score of 1.038 log (see Figure 14 and Figure 15).

Table 13. Best decision rules

DECISION RULE 1	
IF	IPAQ VIGOROUS SCORES t < = 240
AND	Age t > 76
AND	LEISURE ACTIVITIES TOTAL OTHER ACTIVITIES INDEX t < = 2
AND	Years Education t < = 11
THEN	Training Median Performance = .386 (4% of sample)
DECISION RULE 2	
IF	IPAQ VIGOROUS SCORES t < = 240
AND	Age t > 76
AND	LEISURE ACTIVITIES TOTAL OTHER ACTIVITIES INDEX t > = 2
AND	Years Education t > 14
THEN	Training Median Performance = .813 (4% of sample)
DECISION RULE 3	
IF	IPAQ VIGOROUS SCORES t > = 240
THEN	Training Median Performance = 1.038(4% of sample)

Figure 14. All decision rules for the top five predictors of PCT Median Performance

IPAQ: VIGOROUS ACTIVITY SCORE t <= 240 [Ave: 0.68, Effect: -0.129] (16)

Age t <= 76 [Ave: 0.826, Effect: 0.146] => PCT performance > 0.826 (7)

Age t > 76 [Ave: 0.566, Effect: -0.114] (9)

LEISURE ACTIVITIES: OTHER ACTIVITIES INDEX t <= 2 [Ave: 0.46, Effect: -0.107] (5)

Years of Education t <= 11 [Ave: 0.386, Effect: -0.074] => PCT performance > 0.386 (1)

Years of Education t > 11 and Years_Education_t <= 15 [Ave: 0.41, Effect: -0.05] => PCT performance > 0.41 (1)

Years of Education t > 15 and Years_Education_t <= 16 [Ave: 0.63, Effect: 0.17] => PCT performance > 0.63 (1)

Years of Education t > 16 and Years_Education_t <= 18 [Ave: 0.442, Effect: -0.018] => PCT performance > 0.442 (1)

Years of Education t > 18 [Ave: 0.432, Effect: -0.028] => PCT performance > 0.432 (1)

LEISURE ACTIVITIES: OTHER ACTIVITIES INDEX t > 2 [Ave: 0.7, Effect: 0.133] (4)

Years of Education t <= 11 [Ave: 0.628, Effect: -0.071] => PCT performance > 0.628 (1)

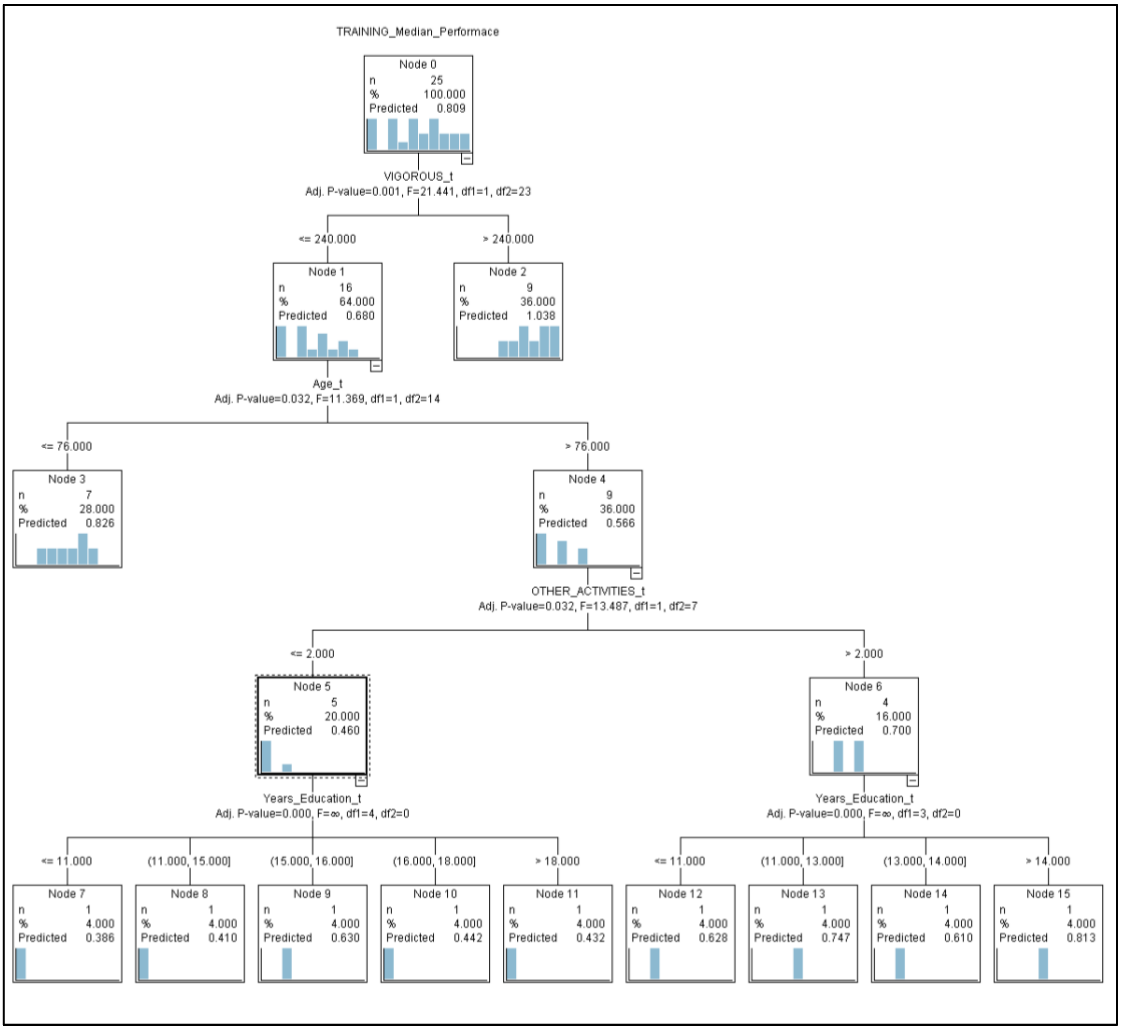
Years of Education t > 11 and Years_Education_t <= 13 [Ave: 0.747, Effect: 0.047] => PCT performance > 0.747 (1)

Years of Education t > 13 and Years_Education_t <= 14 [Ave: 0.61, Effect: -0.09] => PCT performance > 0.61 (1)

Years of Education t > 14 [Ave: 0.813, Effect: 0.114] => PCT performance > 0.813 (1)

IPAQ: VIGOROUS ACTIVITY SCORE t > 240 [Ave: 1.038, Effect: 0.229] => PCT performance > 1.038 (9)

Figure 15. Decision rules tree for the top five predictors of PCT Median Performance



Target: PCT Slope Performance

In Table 14 the CHAID Tree classifier was fit to the *PCT T Slope Performance* scores and resulted in a favourable correlation value of .870 and relative error of .244. The Figure 16 shows the CHAID scatterplot of observed and predicted target values and the ranked most important predictors to build this model: Leisure Activities: Physical Activity Index, Concussion history, Stroke history, D-KEFS VFT Category Fluency and D-KEFS VFT Letter Fluency reveal to be very important features to construction of the tree model (see Figure 17).

Table 14. Target PCT Slope Performance

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	CHAID 1	< 1	0.87	5	0.244

Figure 16. CHAID scatterplot between the observed and predicted values correlation and a measure about the most important predictors

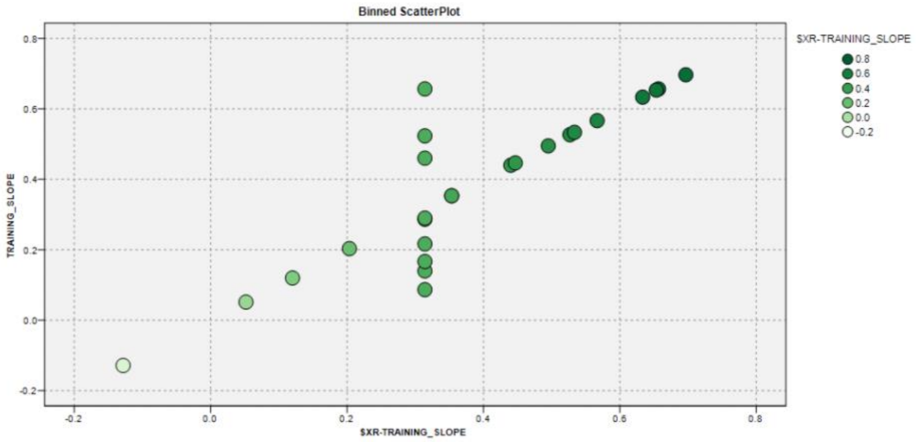
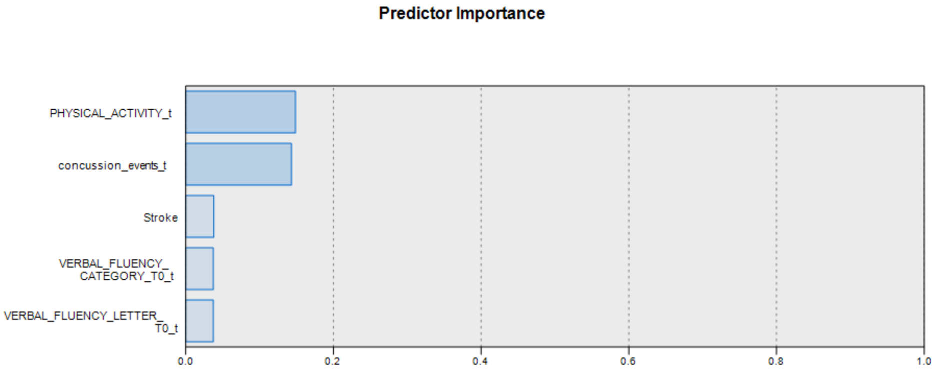


Figure 17. Five top predictors of PCT Slope Performance



The CHAID Tree decision rules are shown in Table 15. For instance, an older adult with low Leisure Activities: Physical Activity Index (≤2) at Age of 69 (or below), will show lower PCT Slope Performance scores of .203 than a senior (Age > 81) with predicted PCT Training Slope scores of .440. Nevertheless, an older adult of AGE of 67 (or higher) with Leisure Activities: Physical Activity Index (scores between 2 and 3), no history of concussion or stroke (see decision rule 3) will be forecasted very high PCT Slope Performance scores of .70 (see Figure 18 and Figure 19).

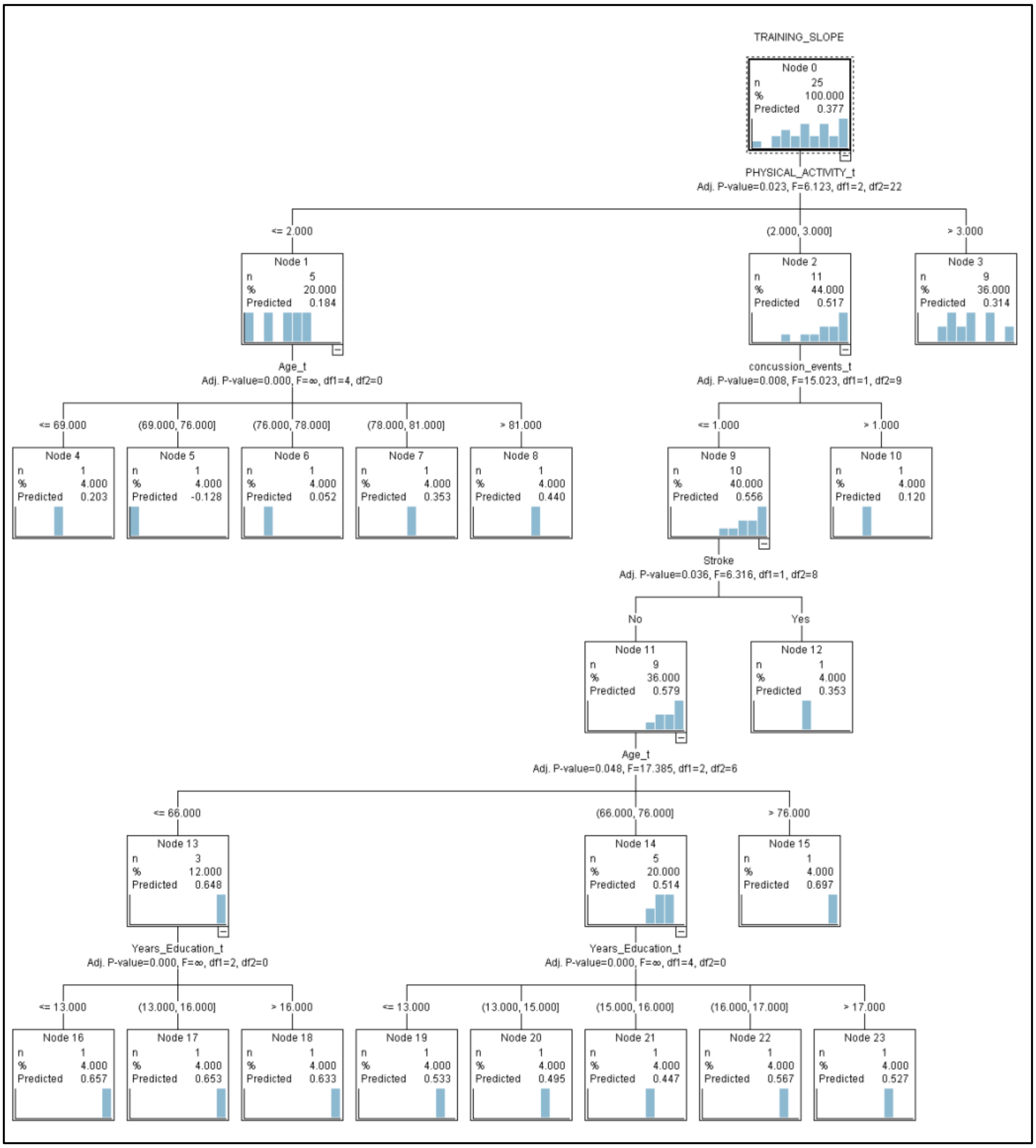
Table 15. Best decision rules

DECISION RULE 1	
IF	LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t <= 2
AND	Age t <= 69
THEN	PCT Training Slope= .203 (4% of sample)
DECISION RULE 2	
IF	LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t <= 2
AND	Age t > 81
THEN	PCT Training Slope= .440 (4% of sample)
DECISION RULE 3	
IF	LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX between 2 and 3
AND	Concussion history t <= 1
AND	Stroke = 0
AND	Age t > 76
THEN	PCT Training Slope = .697 (4% of sample)

Figure 18. All decision rules for the top five predictors of PCT Training Slope

- LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t <= 2 [Ave: 0.184, Effect: -0.193]
  - Age t <= 69 [Ave: 0.203, Effect: 0.019] => PCT performance > 0.203
  - Age t > 69 and Age_t <= 76 [Ave: -0.128, Effect: -0.312] => PCT performance > -0.128
  - Age t > 76 and Age_t <= 78 [Ave: 0.052, Effect: -0.132] => PCT performance > 0.052
  - Age t > 78 and Age_t <= 81 [Ave: 0.353, Effect: 0.169] => PCT performance > 0.353
  - Age t > 81 [Ave: 0.44, Effect: 0.256] => PCT performance > 0.44
- LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t > 2 and t <= 3 [Ave: 0.517, Effect: 0.139]
  - Concussion History t <= 1 [Ave: 0.556, Effect: 0.04]
    - Stroke History = 0 [Ave: 0.579, Effect: 0.023]
      - Age t <= 66 [Ave: 0.648, Effect: 0.069]
        - Years of Education t <= 13 [Ave: 0.657, Effect: 0.009] => PCT performance > 0.657
        - Years of Education t > 13 and Years_Education_t <= 16 [Ave: 0.653, Effect: 0.006] => PCT performance > 0.653
        - Years of Education t > 16 [Ave: 0.633, Effect: -0.014] => PCT performance > 0.633
      - Age t > 66 and Age t <= 76 [Ave: 0.514, Effect: -0.065]
        - Years of Education t <= 13 [Ave: 0.533, Effect: 0.02] => PCT performance > 0.533
        - Years of Education t > 13 and Years_Education_t <= 15 [Ave: 0.495, Effect: -0.019] => PCT performance > 0.495
        - Years of Education t > 15 and Years_Education_t <= 16 [Ave: 0.447, Effect: -0.067] => PCT performance > 0.447
        - Years of Education t > 16 and Years_Education_t <= 17 [Ave: 0.567, Effect: 0.053] => PCT performance > 0.567
        - Years of Education t > 17 [Ave: 0.527, Effect: 0.013] => PCT performance > 0.527
      - Age t > 76 [Ave: 0.697, Effect: 0.118] => PCT performance > 0.697
    - Stroke History = 1 [Ave: 0.353, Effect: -0.203] => PCT performance > 0.353
  - Concussion History t > 1 [Ave: 0.12, Effect: -0.397] => PCT performance > 0.12
- LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t > 3 [Ave: 0.314, Effect: -0.063] => PCT performance > 0.314

Figure 19. Decision rules tree for the top five predictors of PCT Slope Performance



Target: PCT Stability

In Table 16 the CHAID Tree prediction to the *PCT Stability* measure obtained a correlation value of .981 and relative error of .038. The Figure 20 shows the

CHAID scatterplot of observed and predicted target values and the ranked most important predictors to build this model: Leisure Activities: Physical Activity Index, Years of Education, MMSE, D-KEFS VFT Category Fluency, D-KEFS VFT Letter Fluency are very important features (see Figure 21).

Table 16. Target PCT Stability

n	Model	Build Time (mins)	Correlation	No. Fields Used	Relative Error
1	CHAID 1	< 1	0.981	5	0.038

Figure 20. CHAID scatterplot between the observed and predicted values correlation and a measure about the most important predictors

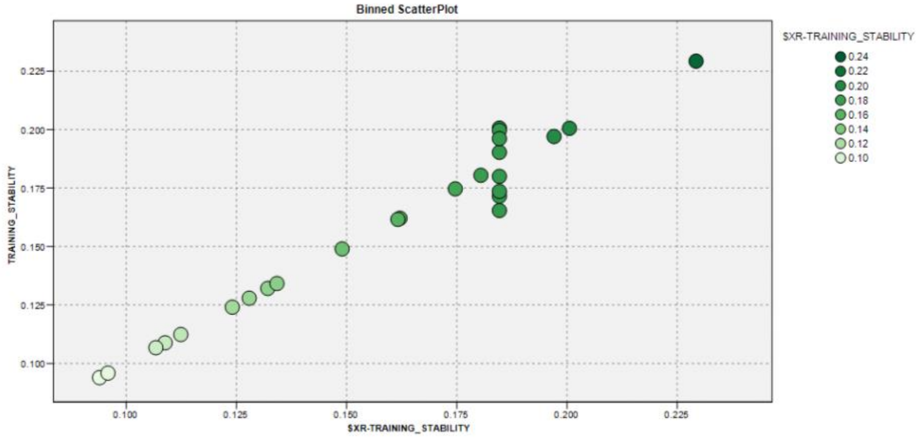
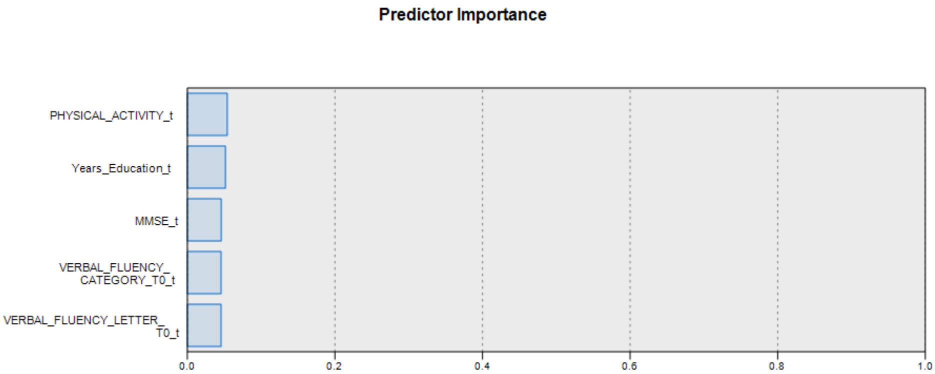


Figure 21. Top five predictors of PCT Stability



Predictor Importance = normalized relative importance from 0 to 1 value

The tree classifier decision rules are shown in Table 17. For instance, an older adult with MMSE ≤ 28 and Years of Education ≤ 11 predicts a PCT Stability as low as .09 points. Instead a senior with MMSE > 29, good physical activity (Leisure Activities: Physical Activity Index > 3) and Year of Education ranging between 18 and 24 has prediction scores on PCT Training Stability of .197. Training variations are clearly in operation (see Figure 22 and Figure 23).

Table 17. Best decision rules

DECISION RULE 1	
IF	MMSE t <= 28
AND	Years of Education t <= 11
THEN	PCT Training Stability = .094 (4% of sample)
DECISION RULE 2	
IF	MMSE t > 29
AND	LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t > 3
AND	Years Education in between t > 18 and t < 24
THEN	PCT Training Stability = .197 (4% of sample)

Figure 22. All decision rules for the top five predictors of PCT Stability

MMSE t <= 28 [Ave: 0.129, Effect: -0.03]

Years of Education t <= 11 [Ave: 0.094, Effect: -0.035] => PCT performance > 0.094

Years of Education t > 11 and t <= 16 [Ave: 0.107, Effect: -0.022] => PCT performance > 0.107

Years of Education t > 16 and t <= 17 [Ave: 0.162, Effect: 0.033] => PCT performance > 0.162

Years of Education t > 17 and t <= 18 [Ave: 0.134, Effect: 0.005] => PCT performance > 0.134

Years of Education t > 18 and t <= 24 [Ave: 0.18, Effect: 0.052] => PCT performance > 0.18

Years of Education t > 24 [Ave: 0.096, Effect: -0.033] => PCT performance > 0.096 |

MMSE t > 28 and t <= 29 [Ave: 0.215, Effect: 0.056]

Gender = 0 [Ave: 0.229, Effect: 0.014] => PCT performance > 0.229

Gender = 1 [Ave: 0.201, Effect: -0.014] => PCT performance > 0.201

MMSE t > 29 [Ave: 0.163, Effect: 0.004]

LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t <= 2 [Ave: 0.141, Effect: -0.022]

Age t <= 76 [Ave: 0.112, Effect: -0.029] => PCT performance > 0.112

Age t > 76 and t <= 81 [Ave: 0.149, Effect: 0.008] => PCT performance > 0.149

Age t > 81 [Ave: 0.162, Effect: 0.021] => PCT performance > 0.162

LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t > 2 and t <= 3 [Ave: 0.185, Effect: 0.022] => PCT performance > 0.185

LEISURE ACTIVITIES: PHYSICAL ACTIVITY INDEX t > 3 [Ave: 0.144, Effect: -0.019]

Years of Education t <= 14 [Ave: 0.175, Effect: 0.031] => PCT performance > 0.175

Years of Education t > 14 and Years_Education_t <= 16 [Ave: 0.124, Effect: -0.02] => PCT performance > 0.124

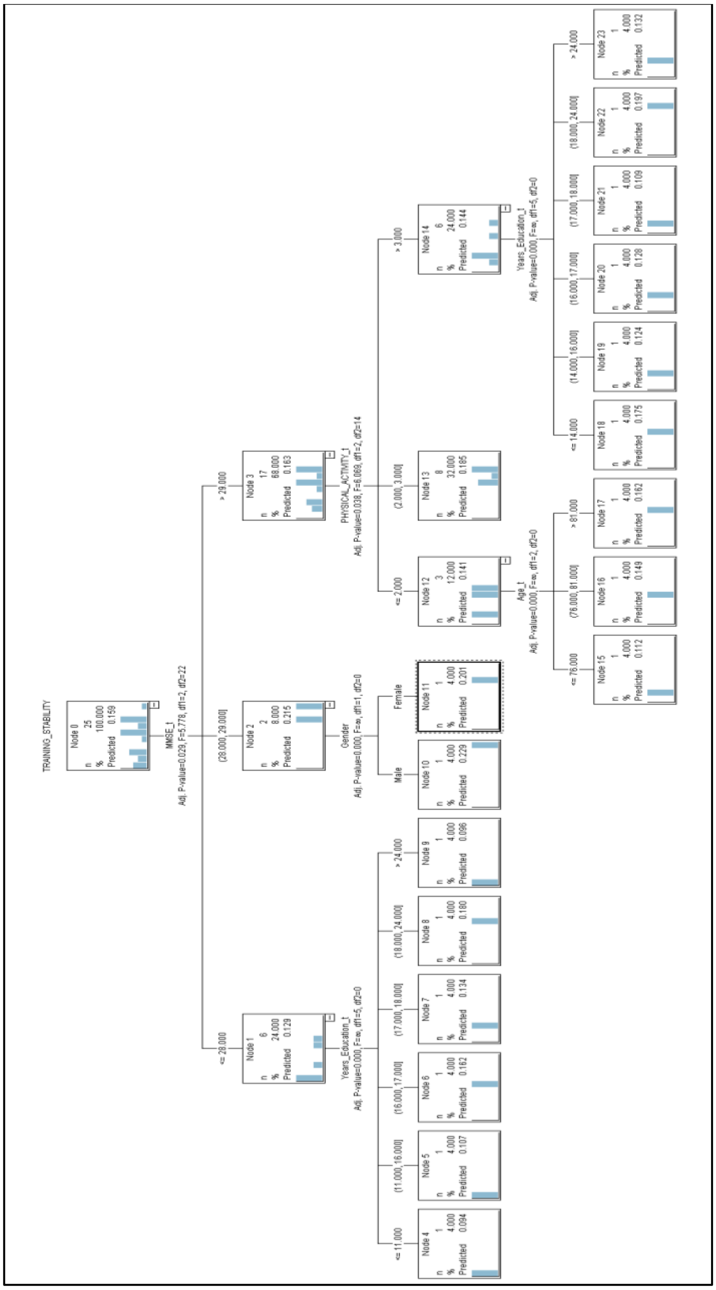
Years of Education t > 16 and Years_Education_t <= 17 [Ave: 0.128, Effect: -0.016] => PCT performance > 0.128

Years of Education t > 17 and Years_Education_t <= 18 [Ave: 0.109, Effect: -0.035] => PCT performance > 0.109

Years of Education t > 18 and Years_Education_t <= 24 [Ave: 0.197, Effect: 0.053] => PCT performance > 0.197

Years of Education t > 24 [Ave: 0.132, Effect: -0.012] => PCT performance > 0.132

Figure 23. Decision rules tree for the top five predictors of PCT Stability



DISCUSSION

The expectations in this dissertation thesis were: 1) to identify if there exists a subsets of feature predictors (i.e. patient's background data, environmental factors, and medical history data available) in order to make reliable predictions on cognitive performance tests scores or on PCT performance at training scores in a group of older adults with SCD; 2) to derive insights (interpretation) about the importance and direction of such sets of predictor features (i.e., the role and impact of each predictor in model). Our findings indicate that AI models can predict the efficacy of Perceptual-Cognitive Training in older adults with SCD. Specifically, AI predictive models identify several subject's profiles with specific features that could alter the PCT or the cognitive test's performance outcome. Thus, the neuropsychologist will be able to know in advance if the patient with SCD meets all the requirements to achieve a certain cognitive test score or a specific PCT performance level. In the contrary case, the neuropsychologist will be able to anticipate to the older adult what he must do in order to achieve that specific performance (e.g. to enhance the physical activity, what kind of physical activity, how often to engage in that physical activity).

First, we looked for AI predictive models that based on an accurate combination of specific predictors (e.g. *Age, Gender, Years of Education, Bilingualism, Concussion History, Stroke History, MMSE, MAC-Q, Leisure Activities: Total Index, Leisure Activities: Social Index, Leisure Activities: Physical Activity Index, Leisure Activities: Mental Index, Leisure Activities: Other Activities Index, IPAQ: Total Score, IPAQ: Vigorous Score, IPAQ: Moderate Score*, *IPAQ: Walking Score*) predict patient's scores on targeted cognitive tests (e.g. *CVLT-II Immediate Recall, CVLT-II Short Delay Recall, CVLT-II Long Delay Recall, D-KEFS Verbal Fluency Letter, D-KEFS Verbal*

Fluency Category Switching). The best prediction models (e.g. CHAID), with a strong correlation (i.e. *r* coefficient value lies between 0.8 and 0.9) and a relative error in the low range (i.e. *relative error* value lies between 0.1 and 0.3), were observed for these 3 cognitive targets: *CVLT-II Immediate Recall*, *CVLT-II Short Delay Recall*, and *D-KEFS VFT Category Switching*. These three cognitive targets were evaluated with two different predictive methods (see results section, page 33).

The first predictive method revealed that the *CVLT-II Immediate Recall* performance is better predicted by five predictors in the following order of importance: *IPAQ: Vigorous score*, *IPAQ Walking score*, *Age*, *MAC-Q score*, *MMSE score*, *IPAQ Walking score*. All these five predictors, where the importance of each one was assessed in combination with the others, are likely to accurately predict the performance on this verbal memory subtest before the administration of the test to the subject. The scatterplot between the observed and predicted values for the *CVLT-II Immediate Recall* indicates a strong correlation. Note that the scatterplot's non ideal shape is due to a very low number of subjects (n = 25) where Machine Learning CHAID model shows a certain inaccuracy in predicting the lowest *CVLT-II Immediate Recall* values. The second predictive method, decision tree, pointed out a gender difference in this verbal memory subtest performance that may be modulated by *physical activity* and *leisure activities*. For example, the women are likely to obtain a high score in *CVLT-II Immediate Recall* only if they are highly engaged in leisure activities. This does not mean that men cannot reach a good performance in this verbal memory subtest. Men are likely to obtain a better performance on *CVLT-II Immediate Recall* only if they engage in vigorous physical activity and in leisure activities. Note, there is a discrepancy between the number of men and women in the examined sample (e.g. F = 16; M = 9).

The Machine Learning models revealed that *CVLT-II Short Delay Recall* performance is better predicted by the following five predictors such as *MAC-Q score, MMSE score, IPAQ: Vigorous score, IPAQ Walking score, IPAQ Total score.* All these five predictors in combination are likely to predict the performance with a strong accuracy of this verbal memory subtest before the administration of it to the subject. The scatterplot between the observed and predicted score for the *CVLT-II Short Delay Recall* indicates a strong correlation. Note that although the correlation between the predicted and observed scores is strong, it must be emphasized the small sample size. Generalization of results beyond the study cannot be confirmed and more research is necessary in this regard. The second method used (i.e. decision tree) pointed out that gender and age may have an impact on *CVLT-II Short Delay Recall* performance, although it may be better predicted by physical activity. For example, an older adult that has low scores on IPAQ Moderate activity and is older than 81 years old may obtain low scores on *CVLT-II Short Delay Recall* memory subtest. On the contrary, if an older adult is male and has high scores on IPAQ Moderate or Walking activities than he/she is more likely to obtain a high score on *CVLT-II Short Delay Recall* memory subtest.

The *D-KEFS VFT: Category Switching*, a verbal processing speed subtest, is better predicted by *Leisure Activities Total Index, Gender, MAC-Q score, IPAQ Walking score*. All these five predictors in combination are likely to predict the performance with a strong accuracy of the *D-KEFS VFT: Category Switching* test before the administration of it to the subject. The scatterplot between the observed and predicted values for this verbal processing speed subtest indicates a strong correlation. Note that due to a small sample size the shape of the scatterplot is not ideal and Machine Learning CHAID model shows a certain inaccuracy in predicting the lowest and highest scores. The decision rules emerged from Machine Learning models shows how *D-KEFS VFT: Category Switching* performance is likely to be triggered by concussion history. For example, an older male with a good performance on MMSE (≥ 29), that reports one or more concussion events over the life, is likely to obtain low scores on *D-KEFS VFT: Category Switching* test. In contrast, if an older male obtains a good performance on MMSE (≥ 29) but does not report concussions over the life he is likely to reach high performance on *D-KEFS VFT: Category Switching.*

The above described predictive findings were not observed for *CVLT-II Long Delay Recall* and on *D-KEFS VFT: Letter Fluency*.

Once this subsets of predictors of cognitive performance tests scores was identified, reliable predictions were made on PCT performance at training scores in this sample of older adults with SCD. Machine Learning models revealed that *PCT Median Performance* is better predicted by five predictors in the following order of importance: *IPAQ: Vigorous score, Age, Years of Education, DKEFS VFT Category Fluency and DKEFS VFT Letter Fluency.* All these five predictors in combination are likely to accurately predict the *PCT Median Performance* of a subject with SCD before the engagement in the training. The scatterplot between the observed and predicted score for the *PCT Median Performance* indicates a strong correlation, although the scatterplot's shape is not ideal (i.e. the Machine Learning CHAID model shows a certain inaccuracy in predicting the lowest *PCT Median scores* due to a small sample size). A very interesting result observed in the decision tree (Figure 15) is that the vigorous physical activity has a major impact on PCT performance in an older adult with SCD. For example, it was noted that a high score at IPAQ Vigorous Activity questionnaire (> 240.000) is a strong predictor of high performance on PCT

(1.039). In addition, another remarkable result is that the age, others leisure activities, and the number of years of education are modulators of the PCT performance prediction in case the older adults is not engaged in vigorous physical activity. Having said that, how these findings might be helpful to the neuropsychologist. For example, if a senior is older than 76 years old, has 11 years of education or less and would like to receive benefits from PCT training, an important suggestion would be to engage himself in others leisure activities and to increase his/her physical activity.

The results on PCT Slope Performance are indicating that the best five predictors, sorted by the order of importance are: *Leisure Activities: Physical Activity Index, Concussion history, Stroke history, D-KEFS VFT Category Fluency,* and *D-KEFS VFT Letter Fluency*. All these five predictors in combination are likely to accurately predict the *PCT Slope Performance* of a subject with SCD before his/her engagement in the training. The scatterplot between the observed and predicted score of *PCT Slope Performance* indicates a strong correlation, although the shape is not ideal (i.e. the Machine Learning CHAID model shows a certain inaccuracy in predicting the lowest and the highest *PCT Slope scores* due to a small sample size). A very interesting result observed in the decision tree (Figure 19) is that the engagement in physical activity plays again an important role in predicting a good performance in PCT for older adults with SCD. For example, it was noted that a high score at Leisure Activities: Physical Activity questionnaire (> 3) is a good predictor of performance on PCT (.314). In case that the older adult with SCD is not engaged in physical activity, then the high age, the presence of concussion history and/or stroke history and low education are negative modulators of the PCT performance prediction. Having said that, for neuropsychologist it would be beneficial to know that the only one patient's feature that can be juggled in order to benefit from the proposed cognitive intervention is again the participation in physical activity.

The remarkable prediction made based on the PCT Stability scores is the enhancement prediction that an older adult with SCD may obtain session after session. For example, if a senior has a performance on MMSE higher than 29 and does moderate physical activity than he/she is likely to increase the PCT performance by .144 scores every time he/she undergoes this cognitive training. In addition, if the older adult with SCD has also a high education than he/she is likely to increase the PCT performance by .197 scores every time he/she undergoes this training.

Age has a great impact on cognitive functioning and is the major risk factor for cognitive decline[65]. In addition, the aging process is by nature accompanied by many changes both in brain structure and brain activity, changes that can ultimately have negative effects on cognitive function[73]. Changes in brain activity that come with aging can also reflect brain plasticity that serves to preserve cognitive functions[71]. The influence of *Age* on cognitive performance is well explained in the literature on aging[166–168]. Therefore, the observed effect of age as a predictor of the cognitive and PCT performance of these older adults with SCD is in line with the available scientific results.

A number of studies have also investigated gender differences in relation to cognitive performance on tasks[169]. It is generally assumed that women perform better on episodic memory tasks[170] and on measures of verbal ability[171,172], while men tend to perform slightly better on tests measuring visuospatial ability such as mental rotation tasks[173]. Gender differences also appear to be stable over time. For example, de Frias, Nilsson, and Herlitz (2006) found that over 10-year period women performed better on tasks measuring semantic fluency, episodic recall, and recognition. These findings provide explanation to the observed *Gender* influence on cognitive performance as a predictor.

A review study of Valenzuela & Sachdev (2006) suggests that higher levels of education, together with other beneficial lifestyle factors, reduces the risk of future dementia. In contrast, a lower level of education has been related to higher risk of dementia[174]. A study of Sorman (2015) on aging reports that educational attainment attenuates cognitive decline among non-demented elderly and is one of the major components that contributes to the cognitive reserve.

Physical activity upregulates many of the biochemical markers in the brain that enhance the capability for cognitive and neural plasticity[175]. For example, the production of brain-derived neurotrophic factor (BDNF), a molecule involved in neurogenesis, neuroprotection, and learning and memory operations, is increased with both physical exercise and mental activity[73,176]. In the short term, an increase in neurotrophins like BDNF can enhance the capacity of existing neurons, increasing synaptic plasticity, augmented prefrontal cortex volume and enhanced executive function performance[177]. The concept of adult neurogenesis also provides an interesting alternative explanation for how engaging in physical and leisure activities can improve the performance in cognitive training and in cognitive tests[176]. Neurogenesis is diminished with age, but exercise reliably reverses the normal decline in neurogenesis and is accompanied by improved Morris water maze performance[178,179].

The concept of "cognitive reserve" provides an interesting alternative explanation for these findings. It is well known that cognitive reserve is theoretically achieved through increasing neuronal or grey matter density[74,180,181]. A longitudinal study of Sorman (2015) explains how demographic factors (e.g. age, gender, education), health (e.g. concussion and stroke events) and lifestyle such as the engagement in leisure activities (e.g. mental and socio-cultural), and in physical activity are positively associated with cognitive functioning. Taken together, a number of reviews also propose that engagement in leisure activities and in constant physical activity can be beneficial in maintaining a good cognitive ability and in boosting the cortical plasticity[167,182–184]. Therefore, the Machine Learning prediction of *physical activity* (e.g. *IPAQ: Vigorous, Moderate, Walking Scores)* and *leisure activities* (e.g. *Leisure Activities Total Index)* as the best predictors of cognitive and PCT performance is in line with the cognitive reserve model.

In conclusion, Machine Learning algorithms have specific and unique characteristics to address the modelling prediction on the examined data. Nevertheless, the results presented are based on the best models selected with higher fit to data and prediction. The key factor indeed is the search for the best predictive models that maximise prediction or interpretation of predictors within "readable" models. Notably the patient patterns identified by the Machine Learning models are useful patient profiles all together that would support on patient screening and intervention decisions. Thus, Machine Learning models in this study inform, drive and ultimately leverage the decision-making of neuropsychologists to weather expose or reject a patient to the PCT intervention.

LIMITATIONS

The first limitation is the small sample size that may limit the generalizations that can be made from these findings to the greater population of older adults. No test set of data has been retained for complete cross-validation. The dataset is too spare to offer a testing and validation phase. Possibly this could be considered the major limitation in the data science application of this thesis. Nevertheless, the results prove how to select some best models with higher fit to data and prediction. Such procedures rely on competition across algorithms or final aggregation of models' outcomes. For this reason, the current study was considered "pilot study" and further research would benefit from employing larger number of participants to increase confidence in the reliability and validity of the findings.

Another limitation related to the small sample size is the disparity in numbers between men and women. *Gender* has emerged several times as a predictor of cognitive and PCT performance, but these results should be replicated by further research to ensure reliability.

It is important also to consider that decision support tools (AI and Machine Learning models) can be beneficial but consistency and bias on the modelling must always be taken into account. For this reason, further research on a larger sample size is required to reduce the bias and to increase the reliability and validity of the findings.

As this was a pilot study, and there is, to our knowledge, limited research on the use of the AI predictive models in clinical setting or healthcare, future research would benefit from assessing the efficacy of such predictions.

CONCLUSIONS

Machine Learning or AI models are useful technologies for the application in the neuropsychology domain, especially in the choice of a personalized treatment. This dessertation findings suggest that the applied Machine Learning models may help to classify and estimate the cognitive performance and the training outcomes in anticipation of exposing a subject to the PCT intervention. Notably the patient patterns identified by the Machine Learning models are useful profils that may facilitate the decision-making capacity of the neuropsychologist to whether an older adult with SCD can achieve a certain cognitive test score or a specific PCT performance level. For example, background variables such as “older age”, “less education”, “gender”, “concussion/stroke history” in an older adult with SCD may be indicative of a low performance on cognitive tests or PCT intervention. Nevertheless, if an older adult with such characteristics engages in moderate or vigorous physical activity or in moderate or vigorous leisure activity, he/she is likely to achieve high scores on cognitive tests or high-performance at PCT.

In conclusion, such predictive modelling outcome provides evidence of the potential of the Machine Learning sensitivity analysis for an efficient personalized intervention using Perceptual-Cognitive Training in older adults with Subjective Cognitive Decline.

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Appendix A

Memory Complaint Questionnaire (MAC-Q)

As compared to when you were in high school or college, how would you describe your ability to perform the following tasks involving your memory?

	Much better now (1)	Somewhat better now (2)	About the same (3)	Somewhat poorer now (4)	Much poorer now (5)
1. Remembering the name of a person just introduced to you					
2. Recalling telephone numbers or zip codes that you use on a daily or weekly basis					
3. Recalling where you have put objects (such as keys) in your home or office					
4. Remembering specific facts from a newspaper or magazine article you have just finished reading					
5. Remembering the item(s) you intended to buy when you arrive at the grocery store or pharmacy					
6. In general, how would you describe your memory as compared to when you were in high school?	(2)	(4)	(6)	(8)	(10)

Total Score ________

Appendix B

Mini-Mental State Examination (MMSE)

Item *Points*

1. What is the:
    - Year?.../1
    - Season?.../1
    - Month?../1
    - Today's date?../1
    - Today's day of the week……………………………………………………………………../1

2. Where are we?
    - Country:../1
    - Province:../1
    - City:../1
    - Name of building:../1
    - Floor of the building…...…………………………………………………………………....../1

3. "*I will say three objects. Repeat them after I have said all three words. Remember what they are because I am going to ask you to name them again in a few minutes.*"
Name the three objects (ball, car, man), taking one second to say each. (Repeat the answers until the patient learns all three, up to six trials. The score is based on the first trial.)
…………………………………………………………………………………………………...../3

4a. Serial 7s. "*Subtract 7 from 100. Then subtract 7 from that number, etc. Stop after 5 subtractions"* (93, 86, 79, 72, 65).
Score the total number of correct answers

4b. (Only do this test if participant struggled with #4) *Spell "WORLD" backwards.* The score is the number of letters in correct order (e.g., DLROW = 5, DLORW = 3)……………………………/5

5. *What are the three words I asked you to remember?* …….……………………………………../3

6. Point to pencil/pen and a wristwatch. Have the patient name them as you point……………..../2

7. *Please repeat after me:* "No ifs, ands, or buts" (only one trial)…………………………………../1

8. Have the patient follow a three-stage command: "*Take the paper in your right/left (non-dominant hand). Fold the paper in half. Put the paper on the floor.*"………………………….……/3

9. Have the patient read and obey the following: "Close your eyes."
"Please read this sentence aloud and then do what it says"………………………………...…../1

10. Have the patient write a complete sentence of his or her own choice……………………….../1

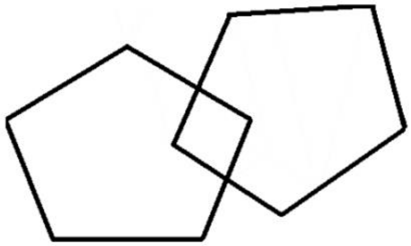
11. Have the patient copy the following design (overlapping pentagons)
"Here is a drawing. Please copy the drawing on the same paper"…………………………………/1

CLOSE YOUR EYES

……

……

……



Appendix C

LEISURE ACTIVITY QUESTIONNAIRE

Daniel Eriksson Sorman et al. (2013)

Participant Number ……………………………

INSTRUCTION

For each activity, please indicate your frequency of participation in the last three months, using the following codes:

0 = NEVER
1 = OCCASIONALLY
2 = A FEW TIMES A MONTH
3 = A FEW TIMES PER WEEK
4 = EVERY DAY

1.	**Travel/Trips**	0	1	2	3	4
2.	**Restaurants visits**	0	1	2	3	4
3.	**Spending time with family, relatives and friends**	0	1	2	3	4
4.	**Attending courses/workshops**	0	1	2	3	4
5.	**Religious assemblies**	0	1	2	3	4
6.	**Volunteering work**	0	1	2	3	4
7.	**Sports/Exercise/Walking in the forest or parks**	0	1	2	3	4
8.	**Hunting/ Fishing**	0	1	2	3	4
9.	**Reading books**	0	1	2	3	4
10	**Reading magazines**	0	1	2	3	4
11	**Reading newspapers**	0	1	2	3	4
12	**Watching TV/ Listening to the radio**	0	1	2	3	4
13	**Movies/Concerts/Theatre**	0	1	2	3	4
14	**Playing a musical instrument**	0	1	2	3	4
15	**Needlework**	0	1	2	3	4
16	**Other hobbies/Activities**	0	1	2	3	4

Appendix D

INTERNATIONAL PHYSICAL ACTIVITY QUESTIONNAIRE (August 2002)

SHORT LAST 7 DAYS SELF-ADMINISTERED FORMAT

FOR USE WITH MIDDLE-AGED ADULTS AND ELDERLY

The International Physical Activity Questionnaires (IPAQ) comprises a set of 4 questionnaires. Long (5 activity domains asked independently) and short (4 generic items) versions for use by either telephone or self-administered methods are available. The purpose of the questionnaires is to provide common instruments that can be used to obtain internationally comparable data on health–related physical activity.

Background on IPAQ

The development of an international measure for physical activity commenced in Geneva in 1998 and was followed by extensive reliability and validity testing undertaken across 12 countries (14 sites) during 2000. The final results suggest that these measures have acceptable measurement properties for use in many settings and in different languages, and are suitable for national population-based prevalence studies of participation in physical activity.

Using IPAQ

Use of the IPAQ instruments for monitoring and research purposes is encouraged. It is recommended that no changes be made to the order or wording of the questions as this will affect the psychometric properties of the instruments.

Translation from English and Cultural Adaptation

Translation from English is supported to facilitate worldwide use of IPAQ. Information on the availability of IPAQ in different languages can be obtained at www.ipaq.ki.se. If a new translation is undertaken we highly recommend using the prescribed back translation methods available on the IPAQ website. If possible please consider making your translated version of IPAQ available to others by contributing it to the IPAQ website. Further details on translation and cultural adaptation can be downloaded from the website.

Further Developments of IPAQ

International collaboration on IPAQ is on-going and an ***International Physical Activity Prevalence Study*** is in progress. For further information see the IPAQ website.

More Information

More detailed information on the IPAQ process and the research methods used in the development of IPAQ instruments is available at www.ipaq.ki.se and Booth, M.L. (2000). *Assessment of Physical Activity: An International Perspective.* Research Quarterly for Exercise and Sport, 71 (2): s114-20. Other scientific publications and presentations on the use of IPAQ are summarized on the website.

INTERNATIONAL PHYSICAL ACTIVITY QUESTIONNAIRE

We are interested in finding out about the kinds of physical activities that people do as part of their everyday lives. The questions will ask you about the time you spent being physically active in the **last 7 days**. Please answer each question even if you do not consider yourself to be an active person. Please think about the activities you do at work, as part of your house and yard work, to get from place to place, and in your spare time for recreation, exercise or sport.

Think about all the **vigorous** activities that you did in the **last 7 days**. **Vigorous** physical activities refer to activities that take hard physical effort and make you breathe much harder than normal. Think *only* about those physical activities that you did for at least 10 minutes at a time.

1. During the **last 7 days**, on how many days did you do **vigorous** physical activities like heavy lifting, digging, aerobics, or fast bicycling?

   _____ **days per week**

   ☐ No vigorous physical activities → ***Skip to question 3***

2. How much time did you usually spend doing **vigorous** physical activities on one of those days?

   _____ **hours per day** _____ **minutes per day**

   ☐ Don't know/Not sure

Think about all the **moderate** activities that you did in the **last 7 days**. **Moderate** activities refer to activities that take moderate physical effort and make you breathe somewhat harder than normal. Think only about those physical activities that you did for at least 10 minutes at a time.

3. During the **last 7 days**, on how many days did you do **moderate** physical activities like carrying light loads, bicycling at a regular pace, or doubles tennis? Do not include walking.

   _____ **days per week**

   ☐ No moderate physical activities → ***Skip to question 5***

4. How much time did you usually spend doing **moderate** physical activities on one of those days?

_____ **hours per day** _____ **minutes per day**

☐ Don't know/Not sure

Think about the time you spent **walking** in the **last 7 days**. This includes at work and at home, walking to travel from place to place, and any other walking that you might do solely for recreation, sport, exercise, or leisure.

5. During the **last 7 days**, on how many days did you **walk** for at least 10 minutes at a time?

_____ **days per week**

☐ No walking → ***Skip to question 7***

6. How much time did you usually spend **walking** on one of those days?

_____ **hours per day** _____ **minutes per day**

☐ Don't know/Not sure

The last question is about the time you spent **sitting** on weekdays during the **last 7 days**. Include time spent at work, at home, while doing course work and during leisure time. This may include time spent sitting at a desk, visiting friends, reading, or sitting or lying down to watch television.

7. During the **last 7 days**, how much time did you spend **sitting** on a **week day**?

_____ **hours per day** _____ **minutes per day**

☐ Don't know/Not sure

This is the end of the questionnaire, thank you for participating.